IMF Governance in a Fragmenting World*

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Abstract

We study how IMF governance should evolve as crisis risk shifts across countries and geopolitical rivalry reshapes international economic ties. Our welfare benchmark links contribution shares and voting weights to economic size, crisis risk, and trade spillovers. Under similar crisis probabilities for all economies, an analytical approximation supports a quota-like rule based on GDP at PPP and openness. A trade model calibrated to 127 countries shows that the post-Bretton-Woods concentration of borrowing among emerging and developing economies instead calls for substantial changes to existing quota shares. Severe fragmentation into US- and China-centered spheres reduces cross-sphere trade substantially but leaves contribution shares almost unchanged. The results suggest that potential institutional tensions would stem primarily from lingering quota misalignment, possibly exacerbated by US–China disagreements over crisis management.

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1 Introduction

How should the governance of the International Monetary Fund adapt when the structure of the world economy changes? The IMF was designed as a quota-based institution with quotas determining contributions, access to resources, voting power, and the allocation of Special Drawing Rights.[1] This design has always mixed an economic logic with a political one. The distribution of crisis risk has changed markedly since the Fund was created. A governance rule suited to a membership made up of countries facing comparable balance-of-payments risks may cease to reflect their welfare stakes once some become recurrent creditors and others the principal borrowers. It may also become politically fragile when economic integration is reorganized around rival geopolitical blocs.

The tension is clearest in current quota debates. Successive reforms have increased the size of Fund resources more easily than they have reallocated formal voice. This matters because the Fund is not just a pool of reserves. It is an institution that decides, under political constraints, whether the resources of all members will be used to stabilize one member and, through that decision, protect the network of trade and financial relationships in which the rest of the membership is embedded. A voting rule is therefore a statement about whose exposure to crises counts, and by how much.

This paper studies IMF governance and how the allocation of voice should adapt as the global and geopolitical environment changes. The normative benchmark is derived from a simple principle: IMF lending decisions should aggregate not only whether member countries favor or oppose a rescue, but also how intensely their citizens are affected by the decision. Following the logic of Barberà and Jackson (2006), the benchmark assigns weights according to expected preference intensities, i.e. a member receives more weight when, conditional on voting for or against a loan, the expected welfare consequences for its citizens are larger.

We quantify these preference intensities using a structural trade model calibrated to 2019 data for 127 countries. A crisis in a given country is modeled as an adverse shock that can be avoided through IMF assistance. The welfare gain to other countries from rescuing the crisis country is then disciplined by the trade equilibrium: the rescue preserves demand for their exports to the crisis country and prevents an increase in the effective cost of intermediate goods imported from it. Aggregating these bilateral welfare gains over the distribution of possible crises determines both vote-specific weights and contribution

[1] Special Drawing Rights (SDRs) are an international reserve asset created by the IMF whose value is based on a basket of major currencies and which countries can exchange for usable foreign currency when needed.

shares. These objects depend not only on a country's own size and crisis probability, but also on the crisis probabilities of its trading partners and on the model-implied geography of trade.

We consider the governance benchmark across three historically relevant regimes. The first is modeled on the period from the Funds creation to the breakdown of the Bretton Woods arrangement. Among its members, the Fund operated like a credit union of economies that could plausibly need liquidity support.[2] We model this period as a *balanced* environment in which member countries face similar probabilities of requiring assistance. Under the maintained assumptions, the analytical approximation makes the gross expected welfare stake increase with GDP at PPP and openness: GDP at PPP captures the scale of real income at stake, while openness captures exposure to foreign crises through trade. Under the common-bias and positive-weight proportionality conditions developed below, this stake implements a quota-like voting benchmark. The quantitative analysis computes the corresponding balanced-risk allocation using the full model.

The second regime is modeled after the period that begins with the end of Bretton Woods and continues through the post-Cold-War enlargement of IMF membership. During this period, the risk environment becomes *unbalanced*. Advanced economies are much less likely to draw on IMF resources, while emerging and developing economies become the main users of Fund lending. The governance benchmark therefore changes. A quota rule based only on size and aggregate openness overweights members that are large but unlikely either to borrow or to be strongly exposed to likely borrowers, and underweights members whose expected welfare gains from crisis lending are large. A quota-like implementation must account for heterogeneity in crisis risk: a country's benchmark rises with its own probability of needing IMF assistance, while the trade component weights bilateral links by the crisis probabilities of trading partners. Solving the full quantitative model for this unbalanced regime yields contribution shares that differ substantially from existing IMF quotas, with a much larger overall mismatch than in the balanced benchmark.

The rise of China, together with the return of industrial policy, sanctions, friend-shoring, and security-motivated trade restrictions, raises the possibility that the relevant counterfactual is no longer a single integrated world economy. We model this third regime as a *bipolar* geoeconomic environment in which countries sort into US- and China-centered economic spheres. Membership in one sphere raises the cost of trading with the rival hegemon and, by a smaller amount, with ordinary members of the rival sphere. We contrast the integrated, unbalanced economy with a hypothetical severe scenario in which the ad-

[2]This was a largely unipolar order: the Soviet Union did not become an IMF member and most Soviet-bloc economies were absent or left early.

ditional cost alone would halve trade with the rival hegemon, holding other determinants fixed. Countries choose the sphere that maximizes their expected aggregate output, accounting for the effects of crises at home and abroad and the benefits and financing costs of IMF intervention, taking other countries' assignments as given. Because each choice changes the markets and suppliers available within the two spheres, the alignment problem can admit multiple equilibria.

Accounting for the post-Bretton-Woods concentration of borrowing among emerging and developing economies substantially widens the gap between existing quotas and the model's benchmark. High-yield countries in our sample hold 16.7 percent of IMF quotas, while their model-implied contribution share rises from 25.7 percent in a balanced scenario to 57.5 percent in the unbalanced world. This reallocation reflects both members' own likelihood of needing assistance and their trade exposure to likely borrowers. Geoeconomic fragmentation produces a 45.4 percent decrease in cross-sphere trade, but no country's contribution share changes by more than 0.06 percentage points relative to the unbalanced benchmark. The counterfactual highlights the underrepresentation of the China-centered sphere relative to the model's contribution benchmark. Its members account for 57.8 percent of model-implied optimal contributions but only 34.7 percent of existing quotas within our sample. Thus, the results suggest that quotas have remained poorly aligned with the distribution of expected gains from Fund lending, and that the additional effect of geoeconomic fragmentation is small compared with the reallocation induced by asymmetric crisis risk, while highlighting the underrepresentation of the China-centered sphere.

These findings suggest that potential tensions within the Fund would arise primarily from the lingering misalignment of quotas with the post-Bretton-Woods distribution of crisis risk and expected gains from assistance. Rivalry between the United States and China could exacerbate that problem through disagreements over how crises should be managed. Such political disagreements are beyond the model; the fragmentation exercise shows that a large reduction in trade between their economic spheres need not itself require a substantial further reallocation of contribution shares.

Related literature. This paper connects four literatures. The first studies IMF governance and quota reform, including the under-representation of emerging and developing economies (Buira, 2005; Cottarelli, 2005; Bird and Rowlands, 2006; Rapkin and Strand, 2003, 2006; Woods, 2005; Frey and Stutzer, 2006) and the politics of institutional inertia (Vestergaard and Wade, 2015; Truman, 2018). Official IMF documents describe the same tension in institutional terms (International Monetary Fund, 2010, 2020, 2023). We con-

tribute a welfare benchmark that maps the externalities from crisis lending into formal voice.

The second studies voting rules in heterogeneous unions. We adapt Barberà and Jackson (2006)'s preference-intensity logic to IMF lending, disciplining intensities with trade spillovers and Fund contributions. The paper also relates to O'Neill and Peleg (2000), Maggi and Morelli (2006), and the political-economy literature on formal and informal power in the Fund (Copelovitch, 2010; Stone, 2011). Gonzalez-Eiras (2009) is a direct predecessor; we add a structural trade model and quantify how the governance benchmark changes as crisis risk becomes asymmetric across members.

The third provides the quantitative trade environment: the Ricardian structure of Eaton and Kortum (2002) and Alvarez and Lucas (2007), the asymmetric trade costs of Waugh (2010), and the sufficient-statistic welfare logic of Arkolakis, Costinot and Rodríguez-Clare (2012). This machinery maps borrower-specific shocks into bilateral welfare spillovers that standard quota formulas do not recover. We calibrate it with BACI trade data (Gaulier and Zignago, 2010), CEPII gravity variables (Conte, Cotterlaz and Mayer, 2022), and Penn World Table factor and income data (Feenstra, Inklaar and Timmer, 2015).

The fragmentation exercise relates to work on trade dependence and geopolitical influence (Hirschman, 1945; Farrell and Newman, 2019; Drezner, Farrell and Newman, 2021; Kleinman, Liu and Redding, 2024), the strategic use of economic relationships and hegemonic structure (Clayton, Maggiori and Schreger, 2026; Broner et al., 2025), and the channels and welfare costs of fragmentation (Aiyar et al., 2023; Góes and Bekkers, 2022; Bolhuis, Chen and Kett, 2023; Gopinath et al., 2025). We use this setting to assess the sensitivity of the post-Bretton-Woods contribution benchmark to severe trade segmentation.

The remainder of the paper proceeds as follows. Section 2 reviews IMF governance and its historical evolution. Section 3 develops the voting benchmark, introduces the trade model used to quantify the underlying payoffs, derives an analytical approximation and discusses the implications of moral hazard. Section 4 presents the data, calibration, and quantitative comparison of the balanced, imbalanced, and fragmented regimes. Its detailed implementation is in the appendix. Section 5 concludes.

2 Governance in the IMF

The IMF was founded in 1944 to provide temporary financial assistance to members facing balance-of-payments difficulties and to support the stability of the international monetary system. It began with 44 members and has since become nearly universal; following Liechtenstein's accession in 2024, the Fund has 191 members (International Monetary Fund, 2024). Members do not have equal formal power. Each member receives basic votes plus quota-based votes, and quotas remain the central institutional object: they determine the maximum financial contribution a member is obliged to provide, are a key determinant of access to IMF resources, largely determine voting power, and govern SDR allocations (International Monetary Fund, 2026b).[3]

IMF decisions are taken under a weighted voting system. The Board of Governors, comprising one governor from each member, is the Fund's highest decision-making body and delegates most powers to the 25-member Executive Board, which conducts the Fund's day-to-day business. Executive Directors are elected by individual members or groups of members and cast the combined votes of the countries they represent. Under Article XII, Section 5(c) of the Articles of Agreement, the default rule is a majority of votes cast; specified decisions instead require 70 or 85 percent of total voting power. Because the United States holds more than 15 percent of total votes, it can block measures subject to the 85 percent threshold (International Monetary Fund, 2026b).

This multifunctional role of quotas is the source of the governance problem. A formula suitable for measuring contributive capacity need not also measure exposure to crises, financing needs, or the externalities that one member's rescue creates for others, yet these determine the extent to which members can gain or lose welfare directly due to Fund decisions. The original Bretton Woods formula was designed in part to produce politically acceptable shares among the founding members rather than to implement an explicit welfare criterion.[4] Subsequent formulas placed weight on GDP, openness, variability, and reserves, but the rationale for the precise variables and coefficients remained opaque. Since the 2008 reform, the calculated quota formula has been

$$CQS_i = \left(0.50Y_i + 0.30O_i + 0.15V_i + 0.05R_i\right)^k,$$

where Y_i is a blend of GDP at market exchange rates and PPP, O_i is the five-year average

[3] Basic votes were a compromise solution intended to reconcile the principle of sovereign equality with the fact of wide power asymmetries among members.

[4] Mikesell (1994) describes how the formula used to assign quotas among the 44 founding members was constructed to reproduce a negotiated allocation.

of current payments plus current receipts, V_i is the variability of current receipts and net capital flows, R_i is reserves, and $k = 0.95$ is a compression factor (International Monetary Fund, 2021). Calculated quota shares are then rescaled to sum to 100.

The use of GDP at both market and PPP exchange rates reflects a compromise between developing countries, which favored the latter, and advanced economies. The compression factor moderates the dispersion of calculated quota shares. Because the revised formula produced only a limited increase in the voice of emerging economies, the reform also tripled basic votes, raising voting shares relative to quota shares for smaller and less developed members.[5] The formula determines relative quota shares; the total quota envelope remains a discretionary outcome of General Quota Reviews. It is transparent, but it does not stem from an analysis of the welfare effects of IMF intervention and thus makes no attempt to optimize economic outcomes.

General Quota Reviews are conducted at least every five years to assess the adequacy and distribution of quotas. An increase may combine equiproportional and selective components: the former preserves existing shares, whereas the latter moves them toward calculated quotas.[6] Recent reviews illustrate the tension between expanding Fund resources and reallocating representation. The 14th Review, agreed in 2010 and implemented in 2016, doubled quotas and shifted quota shares toward dynamic emerging market and developing countries (International Monetary Fund, 2010, 2016). The 15th Review concluded in 2020 with no quota increase (International Monetary Fund, 2020). The 16th Review approved a proposed 50 percent equiproportional increase; as of July 2026, it had not yet become effective. Once effective, it will strengthen permanent resources without changing quota shares (International Monetary Fund, 2023, 2026a).

The Fund's history motivates the first two regimes in the paper. The balanced unipolar regime corresponds to the early Bretton Woods period, when the Fund operated within a US-led monetary order and most Soviet-bloc economies were absent. Members differed in size and openness, but the later division between recurrent creditors and borrowers was much less pronounced. Under the par-value system, balance-of-payments pressure could arise in advanced economies as well as in poorer members, and high-income economies were among the principal users of IMF credit. Neff (1982) reports that industrialized economies accounted for 68 percent of cumulative Fund drawings through 1973. IMF historical accounts similarly document the subsequent shift in lending toward developing

[5]PPP GDP and the compression factor were among the most contested elements of the reform. The Executive Board retained them in the formula for 20 years, subject to subsequent review, and introduced a mechanism that stabilizes basic votes at 5.5% of total votes.

[6]Historically, the equiproportional component has accounted for about 70% of quota increases, contributing to persistence in the distribution of formal power.

and emerging-market economies later in the century.

The imbalanced unipolar regime corresponds to the period after the collapse of Bretton Woods, when the expected beneficiaries of Fund lending shifted toward emerging and developing economies. Generalized currency floating, oil shocks, capital-account globalization, and the debt crises of the 1980s transformed the demand for Fund resources. Advanced economies became much less likely to borrow, while IMF programs increasingly served developing and emerging economies facing balance-of-payments, debt, banking, and sudden-stop crises. Post-Cold-War enlargement made the Fund nearly universal without eliminating the asymmetry between likely lenders and likely borrowers.

As a counterfactual, we consider a prospective division into US-centered and China-centered economic spheres. The exercise keeps the post-Bretton-Woods asymmetry in crisis risk and asks whether a severe reduction in cross-sphere trade would substantially change model-implied contribution shares. It therefore assesses whether the quota mismatch identified in the unbalanced benchmark is sensitive to a large change in the trade network.

3 Model

3.1 An optimal voting rule

Each Fund member is represented by a national delegate who votes on behalf of its citizens.[7] We study a static binary decision: whether to provide financial assistance to a member that requests it.[8]

There are n member countries, heterogeneous in population, wealth, openness, and the likelihood of requiring assistance. Country i has N_i citizens, each of whom derives utility from per capita consumption of the final good, c_i. With some ex-ante probability, one IMF member j experiences a financial or balance-of-payments shock and requests assistance. A state of the world identifies the borrower and the resulting profile of member preferences over whether to provide the loan. Preferences are summarized by $\vec{u}(j) \in \mathcal{R}^n$, where element $u(j)_i \equiv u_{ij}$ is the utility gain to a representative citizen of country i when

[7] This abstracts from the constituency system under which many Executive Directors represent groups of members.

[8] The framework can be extended to simultaneous requests, in which case the choice is whether to assist all affected members, a subset, or none. Multilateral financial institutions also perform functions other than lending. To the extent that those functions do not require large disbursements of scarce resources, they can be governed separately and are orthogonal to the problem studied here.

country j receives the loan:

$$u_{ij} \equiv c_i^{jb} - c_i^{jn}.$$

The superscripts distinguish consumption when country j is bailed out b and from when assistance is denied n.

After the shock, country i votes for the loan if the resulting utility gain is positive and against it if the gain is negative. Let $h_i : \mathcal{R} \longrightarrow \{b, n\}$ map preferences into votes. Thus $h_i(u_{ij}) = b$ when $u_{ij} > 0$, and $h_i(u_{ij}) = n$ when $u_{ij} < 0$.

In the second stage, these votes are aggregated by a voting rule. Let $v : \mathcal{R}^n \longrightarrow \{0, \frac{1}{2}, 1\}$ denote the outcome as a function of $\vec{u}(j)$. A value of 1 approves the loan, 0 denies it, and $\frac{1}{2}$ denotes a tie resolved by a coin toss.

An efficient voting rule is one that maximizes the expected social welfare function among the class of feasible voting rules.[9] The social welfare function is expected total utility, with equal weight assigned to every citizen regardless of country of residence. We therefore consider voting rules that maximize

$$E\left[\sum_i v(\vec{u}(j)) N_i u_{ij}\right],$$

where the expectation is taken over the distribution of shocks, $\mu(\cdot)$. This relies on the assumption that the probability that country j requires financial assistance is independent of bailout decisions; specifically, $v(\cdot)$ does not affect $\mu(\cdot)$.[10]

We begin with the first-best outcome, in which the underlying preferences u_{ij} are publicly observed or perfectly inferred from the state.[11] The decision rule is

$$v^E(u) = \begin{cases} 1 & \text{if} \quad \sum_i N_i u_{ij} > 0, \\ 0 & \text{if} \quad \sum_i N_i u_{ij} < 0, \\ \frac{1}{2} & \text{if} \quad \sum_i N_i u_{ij} = 0. \end{cases} \tag{1}$$

We also require the same vote-counting rule to apply regardless of which country requests assistance.[12] Following Barberà and Jackson (2006), each country receives one weight

[9] These are voting rules that depend only on the information obtained from the votes of the representatives.

[10] The implications of moral hazard within this framework are analyzed in section 3.5.

[11] With no private information, no vote is needed. We also assume, here and below, that a collective decision can be implemented despite opposition from a subset of members. This requires either an external enforcement mechanism or a self-enforcing institutional arrangement; see Maggi and Morelli (2006).

[12] A state-contingent rule would make vote aggregation depend on the identity of the member requiring assistance. Prospective members may regard such borrower-specific treatment as politically unacceptable.

when it votes in favor of assistance and another when it votes against. The positive-vote weight is

$$\omega_i^b = N_i E[u_{ij}|u_{ij} > 0],$$

where the expectation is taken ex ante over borrower states and their welfare consequences for country i. Thus, the weight on country i's positive vote equals its citizens' expected welfare gain conditional on favoring the loan. The weight on a negative vote is

$$\omega_i^n = -N_i E[u_{ij}|u_{ij} < 0].$$

The associated voting rule is

$$v^E(\vec{u}(j)) = \begin{cases} 1 & \text{if} \quad \sum_{i:h_i(\vec{u}(j))=b} \omega_i^b > \sum_{i:h_i(\vec{u}(j))=n} \omega_i^n, \\ 0 & \text{if} \quad \sum_{i:h_i(\vec{u}(j))=b} \omega_i^b < \sum_{i:h_i(\vec{u}(j))=n} \omega_i^n, \\ \frac{1}{2} & \text{if} \quad \sum_{i:h_i(\vec{u}(j))=b} \omega_i^b = \sum_{i:h_i(\vec{u}(j))=n} \omega_i^n. \end{cases} \tag{2}$$

Proposition 1. If preferences are independent across countries then a voting rule is efficient if and only if it is equivalent up to ties to v^E.

The proof is in Barberà and Jackson (2006). Its independence condition is statistical independence across countries' preference intensities. In this setting this will be satisfied as one country's utility does not depend on the votes cast by other countries.[13]

The quantitative exercise uses the vote-specific weights in (2) but treats the approval threshold as an institutional constraint. A proposal passes when the positive vote weight, as a share of the sum of positive and negative vote weights, reaches γ, set to 0.7 in the baseline to reflect existing IMF voting thresholds. As a robustness exercise we set $\gamma = 1/2$; no exact ties arise in the quantitative exercises.

The vote-specific weights ω_i^b and ω_i^n determine how country i's vote is counted when it supports or opposes assistance. The contribution share ω_i^* instead determines the fraction of each bailout's financing cost borne by country i. Contribution shares are determined before a particular crisis occurs and are proportional to members' aggregate expected net gains from approved IMF interventions. They satisfy $\sum_{i\in\mathcal{P}} \omega_i^* = 1$, where $\mathcal{P}$ is the set of participating members; nonparticipants have zero contribution shares.

Bailout requests are prompted by crises represented as a negative shock to the bor-

[13] Formally we will show that $E[u_{ij}|\vec{h}] = E[u_{ij}|h_i]$.

rower's capital stock. Let

$$\Delta_j^F \equiv \left(\frac{r_j K_j + w_j L_j}{p_j}\right)^{\text{pre}} - \left(\frac{r_j K_j + w_j L_j}{p_j}\right)^{\text{shock}} > 0$$

denote the resulting decline in country j's real factor income. Here K_j and L_j are model capital and labor endowments, r_j and w_j their factor prices, and p_j the final-good price; each expression is evaluated in the corresponding equilibrium. If country j receives assistance, country i's aggregate net gain is

$$U_{ij} \equiv N_i u_{ij} = N_i u_{ij}^+ - \delta\, \omega_i^* \Delta_j^F,$$

where u_{ij}^+ is the gross per capita gain to country i from rescuing country j, before financing costs, and δ scales the financing cost $\delta\Delta_j^F$ shared among members.

Let $S \subseteq \mathcal{P}$ be a conjectured set of borrowers whose requests would be approved, and let μ_j denote country j's annual crisis-arrival intensity. Contribution shares satisfy[14]

$$\omega_i^* = \frac{\sum_{j\in S} \mu_j N_i u_{ij}^+ - \delta\, \omega_i^* \sum_{j\in S} \mu_j \Delta_j^F}{\sum_{k\in\mathcal{P}} \sum_{j\in S} \mu_j U_{kj}}, \qquad i \in \mathcal{P}. \tag{3}$$

The numerator is country i's expected gross gain less its expected financing costs; the denominator is participating members' total expected net gain. Contribution shares affect voting payoffs and are therefore determined jointly with vote-specific weights, participation, and bailout approvals. In equilibrium, S consists precisely of the participating borrowers whose requests pass the voting rule under these contribution shares, and vote-specific weights.

3.2 Quantitative trade model

We now give economic content to the bilateral payoff u_{ij} by quantifying how a shock in country j affects citizens of country i. Lending to country j benefits other members through trade linkages, but it also requires them to finance the bailout. We quantify the trade channel using the Waugh (2010) extension of the Eaton and Kortum (2002) Ricardian trade model. The model generates a gravity equation for bilateral trade in which

[14] We consider equilibria with nonnegative expected net gains for participating members and a strictly positive aggregate net gain.

geographic barriers and importer price levels shape trade flows. Waugh (2010) introduces capital and exporter-side trade frictions to account jointly for bilateral trade and cross-country income differences; we retain these asymmetric trade costs. The remainder of this section states the elements needed for our application; Waugh (2010) provides the full derivation.

Each country produces a nontradable final good, c_i, and a continuum of tradable intermediate goods indexed by $x \in [0,1]$. Consumers value only the final good. Individual intermediates $m_i(x)$ are aggregated into q_i with a symmetric Dixit–Stiglitz aggregator. Final goods and intermediate varieties are produced using capital, labor, and the aggregate intermediate good according to the following nested Cobb–Douglas technologies

$$c_i \;=\; \left[k_i^{\alpha} n_i^{1-\alpha}\right]^{\gamma} q_i^{1-\gamma}.$$

$$m_i(x) \;=\; z_i(x)^{-\theta} \left[k_i^{\alpha} n_i^{1-\alpha}\right]^{\beta} q_i^{1-\beta}.$$

$$q_i \;=\; \left(\int_0^1 m_i(x)^{\frac{\eta-1}{\eta}}\, dx\right)^{\frac{\eta}{\eta-1}}.$$

Countries differ in the technology parameter λ_i, which governs comparative advantage in intermediate production. Country i's inverse-productivity draw for intermediate x is distributed according to

$$z_i(x) \sim \text{Exp}(\lambda_i).$$

The common parameter θ governs the dispersion of productivity, so that $z_i(x)^{-\theta}$ follows a Fréchet distribution. Country i is also endowed with labor L_i and capital K_i. The intermediate-goods price index p_{mi}, bilateral expenditure shares D_{ij}, and final-good price p_i satisfy

$$\begin{aligned}
p_{mi} &= \Upsilon_m \left(\sum_{j=1}^{n} \lambda_j \left(r_j^{\alpha\beta}\, w_j^{(1-\alpha)\beta}\, p_{mj}^{1-\beta}\, \tau_{ij}\right)^{-1/\theta}\right)^{-\theta} \\
D_{ij} &= \frac{\lambda_j \left(r_j^{\alpha\beta}\, w_j^{(1-\alpha)\beta}\, p_{mj}^{1-\beta}\, \tau_{ij}\right)^{-1/\theta}}{\sum_{\ell=1}^{n} \lambda_\ell \left(r_\ell^{\alpha\beta}\, w_\ell^{(1-\alpha)\beta}\, p_{m\ell}^{1-\beta}\, \tau_{i\ell}\right)^{-1/\theta}}, \\
p_i &= \Upsilon_c r_i^{\alpha\gamma} w_i^{(1-\alpha)\gamma} p_{mi}^{1-\gamma},
\end{aligned}$$

where w_i is the wage, r_i is the rental rate of capital, $\tau_{ij} \geq 1$ is the iceberg cost of shipping

from goods country j to country i, and Υ_m and Υ_c are functions of structural parameters. The trade share D_{ij} is the fraction of country i's intermediate expenditure allocated to goods produced in country j. Equilibrium requires goods-market clearing and balanced trade in every country.

We model a negative shock in country j as a temporary reduction in its capital endowment to $K_j(1-\epsilon_j)$, where ϵ_j is the proportional size of the shock. We assume that a bailout restores the pre-shock allocation at a resource cost borne by all members.[15]

Solving the equilibrium in each possible borrower state yields the matrix of gross bailout gains u_{ij}^+. Consistent with section 3.1, u_{ij}^+ is the per-worker output gain in country i when country j is bailed out rather than left to absorb the capital shock. In the quantitative implementation, population is identified with the labor endowment, $N_i = L_i$, so $N_i u_{ij}^+$ is the corresponding aggregate output gain. The resource cost of the bailout is proportional to the model-implied decline in the assisted country's real factor income, Δ_j^F, defined in section 3.1. Country i finances this cost in proportion to its contribution share, so the aggregate payoff that enters the bailout decision is

$$U_{ij} = N_i u_{ij}^+ - \delta\, \omega_i^* \Delta_j^F. \tag{4}$$

Where δ scales the resource cost and ω_i^* is the contribution share defined by the fixed point in expected net payoffs from IMF intervention.

Country i votes for lending to country j if (4) is positive and against it if the expression is negative. Since $U_{ij} = N_i u_{ij}$ and $N_i > 0$, this is the same sign rule as in section 3.1. In the numerical closure, vote weights are conditional means over the conjectured approval set S:

$$\omega_i^b(S) = \frac{\sum_{j\in S} \mu_j U_{ij} \mathbf{1}\{U_{ij} > 0\}}{\sum_{j\in S} \mu_j \mathbf{1}\{U_{ij} > 0\}}, \qquad \omega_i^n(S) = -\frac{\sum_{j\in S} \mu_j U_{ij} \mathbf{1}\{U_{ij} < 0\}}{\sum_{j\in S} \mu_j \mathbf{1}\{U_{ij} < 0\}}.$$

An empty conditioning event receives zero weight. These are the aggregate-payoff equivalents of the vote-specific weights in section 3.1.

We first solve for contribution shares ω_i^* and vote-specific weights ω_i^b and ω_i^n under the two crisis-risk regimes introduced in section 2. In the balanced regime, all countries face the same probability of experiencing a shock and requesting assistance. In the imbalanced regime, sovereign ratings divide countries into *investment-grade* and *high-yield* groups, with the former assigned a lower shock probability.

[15] Because the model is static, no intertemporal lending takes place.

3.3 Analytical approximation

To isolate the determinants of the benchmark and conditions under which it can be implemented with quotas, we derive a first-order approximation using the no-capital version of Alvarez and Lucas (2007). Consequently, we instead represent the shock as a reduction in TFP from λ_j to $\lambda_j(1-\epsilon_j)$. Appendix A.1 approximates its effect on country i's wage and price level while holding wages and prices in countries $k \neq i, j$ fixed. This yields

$$N_i u_{ij}^+ = \begin{cases} \dfrac{N_j}{L_j}\dfrac{1-\gamma}{p_j}\dfrac{\theta L_j w_j \left[1+(1+\theta)D_{jj}\right]}{\Delta_j}\epsilon_j, & i=j, \\ \dfrac{N_i}{L_i}\dfrac{1-\gamma}{p_i}\dfrac{\theta^2\left[L_j w_j D_{ji} + (1+\theta)L_i w_i D_{ij}\right]}{\Delta_i \Delta_j}\epsilon_j, & i \neq j, \end{cases} \tag{5}$$

where

$$\Delta_r \equiv \beta(1+D_{rr}) + \theta\left[1-(1-\beta)D_{rr}\right], \qquad r \in \{i,j\}.$$

Equation (5) shows that a country values a bailout of a trading partner because the bailout preserves demand for its exports and prevents an increase in the effective cost of imported intermediates. The approximation is used to interpret these trade-spillover channels rather than to compute the quantitative voting weights, which are obtained from the full numerical capital-shock model. Appendix A.1 shows that it closely matches the off-diagonal numerical spillovers and, for the governance object of interest, the normalized contribution shares and their reallocation between the balanced and imbalanced crisis-risk regimes.

With this specification of u_{ij}^+, country i's payoff does not depend on the profile of votes cast by other countries. This confirms the statistical independence required by proposition 1. Writing $\Delta(w_j L_j)$ for the decline in factor income in the no-capital approximation, the vote weights for a fixed approval set are

$$\begin{aligned} \omega_i^b(S) &= N_i E_\mu[u_{ij}^+ \mid U_{ij} > 0, j \in S] - \delta\omega_i^* E_\mu[\Delta(w_j L_j) \mid U_{ij} > 0, j \in S], \\ \omega_i^n(S) &= -N_i E_\mu[u_{ij}^+ \mid U_{ij} < 0, j \in S] + \delta\omega_i^* E_\mu[\Delta(w_j L_j) \mid U_{ij} < 0, j \in S]. \end{aligned}$$

These expressions, together with (5) and the normalized fixed point (3), characterize the benchmark vote weights. The voting rule reduces to single weighted voting when ω_i^n and ω_i^b is the same across countries, i.e. all countries have a common bias given by $\frac{\omega_i^n}{\omega_i^b} \equiv \gamma$.[16] Sufficient conditions for this are that $Prob[u_{ij} > 0]$, and $\frac{E[u_{ij}^+ | u_{ij}>0]}{E[u_{ij}^+ | u_{ij}<0]}$, be independent of

[16] See Corollary 1 in Barberà and Jackson (2006).

country i.[17] The first condition is that every country in the Fund be ex-ante equally likely to vote in favor of a loan. The second condition is satisfied when the distribution of u_{ij}^{+} is the same across countries. Under these conditions, the optimal voting rule can be characterized as assigning weights

$$g_i \equiv N_i \sum_j \mu(j) u_{ij}^{+},$$

where $\mu(j)$ is the borrower's probability of a crisis. When the common-bias condition holds and ω_i^b is proportional to g_i, a loan is approved when votes in favor exceed the threshold $\gamma \sum_{i=1}^{n} g_i$, with $\gamma \equiv \frac{\chi}{1+\chi}$.

Suppose that $\mu(j)$ is common across countries,[18] that $\Delta_j = \overline{\Delta}$, and that cross-country differences arise through productivity λ_i. Using the proportionality between p_i and Y_i/Y_i^{PPP} and omitting common constants (5) gives

$$g_i = \frac{Y_i^{PPP}}{Y_i \Delta_i} \left[Y_i(1 + (1+\theta)D_{ii}) + \theta \frac{X_i + (1+\theta)M_i}{\overline{\Delta}} \right]$$

Under balanced trade, substituting for Δ_i simplifies this expression to

$$g_i = \frac{Y_i^{PPP}}{\beta(1+D_{ii}) + \theta\left[1 - (1-\beta)D_{ii}\right]} \left[(1 + (1+\theta)D_{ii}) + \theta \frac{(2+\theta)(1-D_{ii})}{\overline{\Delta}} \right].$$

To determine how openness affects the weight, differentiate with respect to the domestic expenditure share D_{ii}.[19] For $\beta - \theta(1-\beta) > 0$, the gross expected stake is larger in more open economies. Thus, under the maintained conditions, equal crisis probabilities yield a quota-like benchmark that increases with GDP at PPP and openness.

This balanced case is a useful approximation to the Bretton Woods membership of the Fund. The relevant institutional universe did not include the Soviet Union and, for most of the period, did not include the Soviet bloc. Within the participating non-Soviet economies, under fixed but adjustable parities, high-income countries were recurrent users of Fund resources, so their own crisis probabilities were not negligible. A voting system

[17] Because financing costs depend on the endogenous contribution share, this common-bias condition must be verified from the resulting weights.

[18] We assume $N_i = L_i$, as in the quantitative model, or, more generally, a common N_i/L_i across countries.

[19] This gives

$$\frac{\partial g_i}{\partial D_{ii}} = -Y_i^{PPP} \frac{\theta(2+\theta)\left[\beta - \theta(1-\beta)\right]\left(1-\overline{D}\right)}{\overline{\Delta}\left\{\beta + \theta + \left[\beta - \theta(1-\beta)\right]D_{ii}\right\}^2}.$$

that weights GDP at PPP and openness therefore accords with the economic environment of the early Fund: large and open members both supplied resources and had a meaningful chance of being direct beneficiaries of liquidity support.

We next consider a post-Bretton-Woods environment in which developed countries, denoted North, are less likely to require assistance than the remaining countries, denoted South. Assume that $Prob[u_{ij} > 0]$ and $\frac{E[u_{ij}^{+}|u_{ij}>0]}{E[u_{ij}^{+}|u_{ij}<0]}$ are constant within each region but may differ across regions. In particular, the own-country term u_{ii}^{+} receives greater probability weight in the South. Let $\chi^N = \omega_i^n/\omega_i^b$ for Northern countries and $\chi^S = \omega_i^n/\omega_i^b$ for Southern countries. The benchmark assigns

$$g_i = \frac{Y_i^{PPP}}{Y_i \Delta_i}\left[\mu_i Y_i(1+(1+\theta)D_{ii}) + \theta\left(\mu_S \frac{X_{iS}+(1+\theta)M_{iS}}{\overline{\Delta}^S} + \mu_N \frac{X_{iN}+(1+\theta)M_{iN}}{\overline{\Delta}^N}\right)\right],$$

where $\mu_S > \mu_N$ are the borrower-state weights in the South and North, respectively, and μ_i takes the corresponding regional value. X_{iS} and X_{iN} denote country i's exports to the two regions, with M_{iS} and M_{iN} defined analogously. Because openness may vary systematically with development, average $\overline{\Delta}$ is allowed to differ across regions. A loan to country j is approved when

$$\sum_{u_{ij}>0} \gamma^i g_i > \sum_{u_{ij}<0} g_i, \tag{6}$$

where $\gamma^i = \gamma^S$ (or γ^N) if $i \in$ South (or North) and $\gamma^i \equiv \frac{\chi^i}{1+\chi^i}$. A simpler implementation in some states uses separate votes within the North and South. Northern members favor a loan to country j when their weighted support exceeds $\gamma^N \sum_{i\in N} g_i$, with an analogous threshold in the South. Agreement between the two regional votes determines the Fund's decision; disagreement is resolved by (6).

3.4 Geoeconomic fragmentation

We now use the approximation to characterize the economic forces behind geoeconomic fragmentation between spheres centered around the United States and China, denoted by U and C. Each non-hegemon country i chooses an alliance $a_i \in \{U, C\}$.[20] An analytical rule, (7) below, is used only to clarify first-order alignment incentives. It does not generate the numerical assignment. The reported assignment is instead certified using the numerical full model and the expected-output criterion defined in section A.4; IMF

[20]We are thus assuming that the role of nonaligned countries that can act as connectors between the hegemons is limited. For an analysis of early signs of decoupling between the United States and China that highlights the role of nonaligned countries, see Gopinath et al. (2025).

weights are recomputed under its augmented trade-cost matrix.

If the countries joining the two spheres differ in openness, crisis risk, and exposure to likely borrowers, aggregation by sphere can reveal a substantial difference between formal IMF shares and the model-implied stake in lending. Section 4.2 shows that China receives the largest country contribution weight before and after fragmentation, and that the China sphere contains many other countries with model-implied weights that are much higher than their existing IMF quotas. The main quantitative result is therefore about the distribution of ω^* across the two coalitions.

Appendix A.2 derives an analytical approximation to the country-level alignment rule. Under the assumption that countries' additional trade cost from choosing a sphere are symmetric $\tau_C = \tau_U$ and evaluating first-order effects at the unfragmented baseline, country i aligns with the United States if

$$\begin{aligned}(1+\theta)L_i w_i \left(D_{iUS} + \zeta \sum_{k \in U} D_{ik} \right) + L_{US} w_{US} D_{USi} + \zeta \sum_{k \in U} L_k w_k D_{ki} > \\ (1+\theta)L_i w_i \left(D_{iCH} + \zeta \sum_{k \in C} D_{ik} \right) + L_{CH} w_{CH} D_{CHi} + \zeta \sum_{k \in C} L_k w_k D_{ki} \end{aligned} \tag{7}$$

where $0 \leq \zeta \leq 1$ is the proportional increase in trade costs, relative to the hegemon, between countries of different spheres. Country i is assumed to be small, so its alignment affects only its own wage, intermediate-goods price index, and trade shares. The choice depends on a weighted measure of trade with each sphere, with imports receiving greater weight than exports. Section A.4 describes the numerical implementation.

The alignment vector need not be unique. When ζ is high, the cost of trading with ordinary members of the rival sphere is close to the cost of trading with the rival hegemon itself. Then a country's best response depends on the sphere choices of other countries: if more countries join China, the effective market and supplier base available inside the China sphere expands, while joining the US sphere exposes the country to higher costs on a larger set of China-aligned partners. The same logic works in reverse. For sufficiently strong cross-sphere wedges, this strategic complementarity can generate multiple equilibria, including allocations in which most countries coordinate on the US sphere and allocations in which most countries coordinate on the China sphere. When $\zeta = 0$, by contrast, the choices of non-hegemon countries create no additional bilateral wedge, so the alignment decision is essentially bilateral vis-à-vis the two hegemons and these complementarities are absent or weak.

This structure is consistent with Broner et al. (2025), where policy alignment and

hegemonic structure can generate globalization, fragmentation, and multiplicity of equilibria. It is also consistent with Clayton, Maggiori and Schreger (2026), who show that hegemons optimally coordinate threats across economic relationships; such broad threats make an individual country's payoff depend on how many other relationships fall inside each bloc. Our argument does not require solving an optimal-fragmentation problem. The quantitative exercise verifies every unilateral deviation from the reported assignment under full expected output and asks how model-implied IMF contribution weights are distributed once alignment reshapes trade costs and crisis spillovers.

Given an alignment vector a, the model determines counterfactual trade shares under the augmented costs. The resulting discrepancy between existing quotas and optimal contributions exposes an evident tension for members of the under-weighted sphere. Institutional exit or competition from regional lenders would require an explicit outside option and lies beyond the present exercise.[21]

3.5 Moral hazard

IMF governance can affect welfare through both the resolution of crises and the incentives to prevent them. When domestic policies influence crisis risk, the prospect of assistance may weaken governments' willingness to undertake costly preventive measures. Lending that is desirable once a crisis occurs can therefore have adverse effects on policy choices made beforehand. This consideration matters for our benchmark as reforms that better reflect members' expected gains from assistance may also change their expectations of receiving support. Evaluating governance arrangements then requires considering how the rules shape the distribution of crisis risk itself.

We explore this incentive channel through a parsimonious extension of the framework. A binary domestic policy choice affects the probability of a crisis, while the ruler's private benefits and the consequences of being denied assistance summarize the political incentives governing that choice. Holding voting weights fixed isolates how the approval threshold affects rescue expectations and preventive effort. The exercise illustrates the tension between crisis stabilization and policy discipline with minimal additional structure. It also clarifies why a welfare benchmark derived for a given distribution of shocks need not characterize optimal governance when that distribution responds to the rules.

Suppose that a country's shock probability depends on domestic policy. The ruler

[21] Desai and Vreeland (2011) discusses how declining confidence in the IMF may encourage a shift toward regional monetary arrangements when crisis spillovers, trade exposure, and political trust are stronger within regions than globally.

of country i receives a private benefit $B_i > 0$ from not exerting effort to pursue "good" policies that reduce the likelihood of a financial shock or increase the likelihood of a productive investment opportunity. If a shock occurs and the Fund denies assistance, the ruler incurs a cost $C_i > 0$. The effort decision therefore depends on the probability of receiving a loan under the voting rule. Let $\Delta\mu$ denote the increase in the shock probability when effort is not exerted. The ruler adopts good policies if

$$\frac{B_i}{C_i} < \Delta\mu(1 - s_i) \tag{8}$$

where s_i is the probability of receiving outside finance:

$$s_i = Prob\left[\sum_{k:U_{ki}>0} \omega_k^b \geq \gamma\left(\sum_{k:U_{ki}>0} \omega_k^b + \sum_{k:U_{ki}<0} \omega_k^n\right)\right]. \tag{9}$$

Holding vote weights fixed, a higher threshold γ weakly lowers the probability of rescue in (9) and strengthens the effort incentive in (8). The effect on effort is concentrated among countries close to the incentive threshold in (8).

The jointly optimal characterization of threshold and voting weights; would require an explicit institutional objective that trades off crisis stabilization against effort. Thus, the preceding analysis is indicative. In particular it shows that heterogeneity in B_i/C_i implies that an increase in the voting threshold may discipline some countries without eliminating moral hazard everywhere.

4 Quantitative analysis

4.1 Data and calibration

The model is calibrated for 127 countries in 2019. Appendix table 2 reports the full sample.

Data. Solving the model requires three sets of data: (i) trade flows and gravity variables; (ii) factor endowments and initial real GDP per capita; and (iii) sovereign credit ratings.

Trade shares are constructed from the BACI database on bilateral trade flows from CEPII (Gaulier and Zignago (2010)). Because we model aggregate trade, these shares are measured relative to aggregate GDP. Gravity variables are taken from the CEPII Gravity database (Conte, Cotterlaz and Mayer (2022)), which provides city-weighted bilateral

distances and an indicator for a shared border.

Real GDP per capita, used to initialize λ_i, and capital and labor endowments come from the Penn World Table. Sovereign ratings classify countries as *investment grade* or *high-yield*. We combine ratings from S&P, Moody's, and Morningstar DBRS and classify a country as *high-yield* whenever the agencies disagree.

Parameters. Following Waugh (2010), we set the capital share to $\alpha = 1/3$ and the value-added share in intermediate production to $\beta = 1/3$. The final-good value-added share is $\gamma = 0.75$, as in Alvarez and Lucas (2007), and the dispersion of goods-level productivity is governed by $\theta = 0.1818$. The baseline capital shock is $\epsilon = 0.10$. A bailout requires 70 percent of weighted votes in the baseline; Appendix tables 11 and 14 report the corresponding results under the 50 percent benchmark.

Trade costs and productivity. We estimate trade costs and productivity parameters using the procedure in Waugh (2010).

The Eaton and Kortum (2002) structure implies a ratio representation of bilateral trade shares that eliminates the importer-specific normalization:

$$\log\left(\frac{D_{ij}}{D_{ii}}\right) = S_j - S_i - \frac{1}{\theta}\log\tau_{ij} + \varepsilon_{ij}, \tag{10}$$

where

$$S_i \equiv \log\left[r_i^{-\frac{\alpha\beta}{\theta}}\, w_i^{-\frac{(1-\alpha)\beta}{\theta}}\, p_{mi}^{-\frac{(1-\beta)}{\theta}}\, \lambda_i\right], \tag{11}$$

Following Waugh (2010), we allow exporter-side frictions to make trade costs asymmetric, so $\tau_{ij} \neq \tau_{ji}$ in general. This specification captures systematic barriers faced by exporters in poorer economies and helps match bilateral trade, tradables prices, and cross-country incomes. The symmetric component depends on K distance bins $\mathbf{1}\{\text{dist}_{ij} \in k\}$ and a common-border indicator $\mathbf{1}\{\text{border}_{ij} = 1\}$:

$$\log\tau_{ij} = \sum_k d_k\, \mathbf{1}\{\text{dist}_{ij} \in k\} \;+\; \beta_B\, \mathbf{1}\{\text{border}_{ij} = 1\} \;+\; \varphi_j \;+\; \xi_{ij}. \tag{12}$$

Here φ_j is the exporter-side friction and ξ_{ij} is a bilateral residual. For a given θ, we regress $\log(D_{ij}/D_{ii})$ on importer and exporter fixed effects, which absorb S_i and S_j, and on the cost shifters in (12). The regression yields $\widehat{S}_i$, $\widehat{\varphi}_j$, and fitted trade costs $\widehat{\tau}_{ij}$.

Given $\widehat{\tau}_{ij}$ and $\widehat{S}_i$, the model's price index implies

$$\widehat{p}_{mi} \;=\; \Upsilon_m \left(\sum_j \exp(\widehat{S}_j)\, \widehat{\tau}_{ij}^{-1/\theta} \right)^{-\theta}, \tag{13}$$

where Υ_m is common across countries. Solving the equilibrium system gives wages w_i and rental rates r_i. We then recover λ_i by inverting (11):

$$\lambda_i = \frac{\exp(\widehat{S}_i)}{r_i^{-\frac{\alpha\beta}{\theta}}\, w_i^{-\frac{(1-\alpha)\beta}{\theta}}\, \widehat{p}_{mi}^{-\frac{(1-\beta)}{\theta}}}. \tag{14}$$

Bailout cost. The remaining parameter, δ, scales the cost borne by contributing members as a share of the model-implied decline in the assisted country's real factor income. We discipline this reduced-form cost with voluntary member contributions to the Poverty Reduction and Growth Trust (PRGT) Subsidy Accounts. Specifically, we divide annual contributions by total outstanding IMF credit and average the ratio from 2013 to 2018, the period leading up to the baseline calibration year; this gives $\delta = 0.45\%$.[22]

The ratio varies substantially across years. As a robustness check, we also solve the model using the 2013 value $\delta = 0.58\%$, the second-highest in the 2013–2018 calibration window. Even under this higher calibration, more than half of members are bailed out in both unfragmented benchmarks. The weights of large members change negligibly because the countries at the margin of bailout approval are small in the aggregate.

Crisis probabilities. We calibrate relative request rates from IMF lending arrangements approved between 2000 and 2025 for the 127 sample countries. Using end-2015 sovereign ratings, 54 countries are classified as investment grade (IG) and 73 as high-yield (HY).[23] IG sovereigns received 58 new arrangements in 1,404 country-years, an annual frequency of 4.1 percent; HY sovereigns received 311 in 1,898 country-years, a frequency of 16.4 percent. We therefore set $\mu_N = 0.04$ and $\mu_S = 0.16$ in the baseline.

These are country-specific annual arrival intensities, not a probability distribution over one mutually exclusive borrower state. Governance is evaluated request by request, and

[22] Outstanding credit is total IMF credit and loans to members reported in the annual Financial Statements, including GRA credit, SDA/SAF and Trust Fund loans, and PRG Trust loans. Contributions are annual flows to the PRG Trust Subsidy Accounts. This calibration does not imply that quota subscriptions finance individual programs in proportion to ω_i^*.

[23] We count all approved lending arrangements, including precautionary facilities even when subsequently undrawn, and classify countries unrated by S&P, Moody's, and Morningstar DBRS as high-yield. Every sample country contributes the full 26 country-years. End-2015 ratings capture structural creditworthiness over the observation window rather than a 2019 rating that may respond to the events being counted.

annual expected payoffs add across requests; this treatment permits multiple requests and abstracts from interactions among simultaneous crises. A common rescaling cancels from the sign-conditional formulas for ω_i^b and ω_i^n, but not from the level of expected IMF gains in the assignment criterion. The fragmented equilibrium is therefore conditional on the level of the crisis probabilities.

Fragmentation trade costs. The fragmented regime is designed to contrast the unbalanced economy without additional trade barriers with a severely divided world. We take from Broner et al. (2026) the relative strength of direct and indirect alignment effects, while choosing the overall severity of fragmentation.[24] Their table 2, column 6, reports a coefficient of 0.104 on log bilateral treaty counts. Table 5, column 1, reports a coefficient of 0.020 on the product of the two trading partners' log treaty counts with the United States. Holding one partner at 47 treaties, the upper-quartile value in their comparison, a change in the other partner's treaties from 25 to 47 implies direct and indirect log-trade changes of $0.104 \log(47/25)$ and $0.020 \log(47) \log(47/25)$. We retain their ratio,

$$\zeta = \frac{0.020 \log(47) \log(47/25)}{0.104 \log(47/25)} = \frac{0.020 \log(47)}{0.104} = 0.74.$$

Thus the empirical estimates determine the relative log-trade response, not the magnitude of the fragmentation scenario.[25]

4.2 Quantitative results

We compare three regimes using the same 2019 economic fundamentals. The balanced benchmark assigns all countries a crisis-arrival intensity of 0.16. The imbalanced benchmark retains this intensity for high-yield sovereigns and lowers it to 0.04 for investment-grade sovereigns. The fragmented regime retains these asymmetric crisis intensities and introduces severe trade barriers across endogenously determined US- and China-centered spheres. The first comparison isolates the effect of heterogeneous crisis risk; the second examines how trade fragmentation changes contribution shares and reveals the distribution of the quota mismatch across economic spheres.

[24] We use the September 2026 revision of the Broner et al. (2026) working paper.

[25] The two coefficients come from separate gravity specifications. We use their ratio as a calibration input, not as a causal estimate of the effects of a geopolitical division. The treaty change is for one partner, holding the other's alignment fixed.

4.2.1 Balanced and post-Bretton-Woods benchmarks

Table 1 reports normalized contribution shares for the G20 sovereign members and Spain, a permanent invitee,[26] together with current IMF quota shares renormalized over the 127-country model sample. The balanced regime applies homogeneous crisis risk to 2019 fundamentals; it is a stylized comparison rather than a reconstruction of the Bretton Woods economy. It isolates the allocation implied by economic size and international spillovers when all members are equally likely to request assistance.

The actual to model-implied quota mismatch is visible in the balanced benchmark and becomes substantially larger under post-Bretton-Woods crisis-risk asymmetry. The 73 high-yield countries hold 16.72 percent of current sample-normalized quotas, 25.74 percent of the balanced-model ω^*, and 57.52 percent of unbalanced-model ω^*. Lowering only the investment-grade arrival intensity therefore reallocates 31.8 percentage points of contribution shares toward the high-yield group. Crisis-risk asymmetry thus amplifies and reallocates a discrepancy that already exists under balanced risk. The quota allocation is far from the model's benchmark for the post-Bretton-Woods world, even before fragmentation is introduced.

At the country level, China receives the largest model-implied contribution share in both crisis-risk benchmarks. Its share is 24.91 percent in the balanced benchmark, compared with a sample-normalized quota share of 6.67 percent. The corresponding US shares are 8.69 (in the balanced model) and 18.17 percent (in existing IMF quotas). Introducing post-Bretton-Woods asymmetry lowers the contribution shares of China and the United States to 14.32 and 4.93 percent, respectively, while raising those of high-yield economies that are more likely to borrow and of members exposed to them. Among the G20 economies, the largest increases are for Russia, Indonesia, Brazil, Turkey, and Argentina. The imbalanced allocation therefore differs from the balanced allocation not simply because likely borrowers receive more weight, but because their crises have different spillovers across trading partners.

Appendix table 2 reports the complete country-level results.

4.2.2 Contribution shares under severe trade fragmentation

The counterfactual imposes higher trade costs across US- and China-centered spheres while retaining the imbalanced crisis intensities. Countries choose their sphere according to expected aggregate output, including crisis losses and net gains from IMF intervention.

[26] Spain has been a permanent invitee to G20 Leaders' summits since the inaugural summit in November 2008, attending every subsequent summit. We include it alongside the nineteen sovereign G20 members.

Table 1: Current IMF Quota Shares and Model-Implied Contribution Shares

Country	ISO	Cat.	IMF quota	Balanced	Imbalanced
Argentina	ARG	HY	0.70	0.87	1.97
Australia	AUS	IG	1.44	0.86	0.48
Brazil	BRA	HY	2.42	3.22	7.25
Canada	CAN	IG	2.41	1.21	0.68
China	CHN	IG	6.67	24.91	14.32
France	FRA	IG	4.41	2.12	1.20
Germany	DEU	IG	5.83	3.26	1.86
India	IND	IG	2.87	8.42	4.85
Indonesia	IDN	HY	1.02	3.50	7.79
Italy	ITA	IG	3.30	1.72	0.98
Japan	JPN	IG	6.75	3.89	2.20
Korea	KOR	IG	1.88	1.64	0.93
Mexico	MEX	IG	1.95	1.96	1.11
Russian Federation	RUS	HY	2.82	3.89	8.71
Saudi Arabia	SAU	IG	2.19	1.01	0.57
South Africa	ZAF	IG	0.67	0.44	0.25
Spain	ESP	IG	2.09	1.11	0.65
Turkey	TUR	HY	1.02	0.91	2.02
United Kingdom	GBR	IG	4.41	1.86	1.06
United States	USA	IG	18.17	8.69	4.93

Notes: G20 countries plus Spain. Model columns report normalized ω_i^* contribution shares, in percent, for the model calibrated to 2019. The voting threshold is 70 percent in both model columns. Current IMF quota shares are from the IMF member table updated on July 28, 2026, and are renormalized over all 127 model countries, not over the countries displayed. Source: International Monetary Fund (2026b).

In the reported locally stable assignment, 75 countries join the US sphere and 52 the China sphere. Nominal cross-sphere trade falls by 45.39 percent relative to the unfragmented economy.

This large trade reallocation has only a small additional effect on contribution shares. The high-yield group's combined share changes from 57.52 to 57.59 percent. Across all 127 countries, the largest change is China's decline of 0.060 percentage points; the mean absolute change is 0.0016 points. Expected benefits from a country's own rescue help explain this stability: they account for 95.53 percent of China's gross expected benefit and 97.65 percent of the US gross expected benefit, leaving foreign spillovers a relatively small component for these two countries. Appendix A.5 reports the detailed results, including comparisons that hold sphere membership fixed across regimes.

Within this calibration, the effect of even severe trade fragmentation on the contribution benchmark is secondary to the effect of asymmetric crisis risk. The results point to

the persistent misalignment of quotas with the post-Bretton-Woods system as a potential source of institutional tension. Disagreements between the United States and China over how to administer crisis assistance could aggravate that problem, although such political conflicts are outside the model.

4.2.3 Robustness and scope

Appendix tables 10 and 11 report two variations for the unipolar benchmarks. First, we set $\delta = 0.58\%$, the 2013 value and the second-highest in the 2013–2018 calibration window, which raises the cost to members of providing a bailout. Second, we remove the calibrated supermajority by setting $\gamma = 1/2$, which applies the general voting rule in (2) up to ties.

At $\delta = 0.58\%$, the balanced regime continues to approve 125 borrower states and its contribution shares are unchanged at reported precision. The imbalanced regime approves 125 rather than 126 states because Guinea-Bissau no longer passes. All countries continue to participate. The largest absolute change in an imbalanced contribution share is 0.0114 percentage points, and the mean absolute change is 0.0002 points; the G20 shares in table 1 are unchanged at two decimal places.

Under the 50 percent rule, the balanced model approves 126 of the 127 borrower states and the imbalanced model approves all 127, one additional bailout in each regime relative to the baseline. All countries continue to participate. In the fragmentation extension, the lower threshold brings The Gambia into the Fund and all 127 borrower states receive assistance, with little change in contribution shares. The same sphere assignment remains locally stable after a complete unilateral-deviation check. Appendix table 14 reports the country-level comparison. No exact tie arises in these calculations.

Across these exercises, the relative ranking and contribution shares of the largest economies change little. The principal reallocation continues to arise from moving between balanced and imbalanced crisis intensities: ω_i^* depends on a country's own likelihood of requesting assistance, the likelihoods faced by its trading partners, and the strength of those trade links.

5 Conclusions

IMF governance should reflect members' expected welfare stakes in crisis lending. Our benchmark relates contribution shares and vote-specific weights to economic size, crisis risk, and trade spillovers. When members face similar crisis probabilities, an analytical

approximation supports a quota-like rule based on GDP at PPP and openness, subject to the stated implementation conditions.

The post-Bretton-Woods concentration of borrowing among emerging and developing economies changes this benchmark substantially. Likely borrowers and countries exposed to their crises through trade gain weight. In the quantitative model, high-yield countries account for 57.5 percent of contribution shares but hold only 16.7 percent of sample-normalized IMF quotas. The main governance problem is the lingering mismatch between the allocation of quotas and the distribution of expected gains from Fund assistance.

The fragmentation exercise supports the persistence of this finding. A 45.4 percent fall in nominal trade across US- and China-centered spheres changes individual contribution shares by at most 0.060 percentage points. Within this calibration, trade segmentation has only a small additional effect on the quota benchmark. Potential tensions within the organisation would therefore be more plausibly associated with the unresolved post-Bretton-Woods misalignment, possibly exacerbated by US–China disagreements over how to administer crises. The latter channel requires an explicit account of crisis-management conflict beyond the present model.

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A Appendix

A.1 Approximate welfare effects of a bailout

The full quantitative exercise computes the welfare gains from a bailout by resolving the full n-country equilibrium after a capital shock. This appendix derives a first-order approximation to the same object using a TFP shock in the no-capital model. The purpose of the approximation is to make transparent the main forces behind the quantitative results: a crisis in country j affects its trading partners through the value of their bilateral exports to j and through the cost of the intermediates they import from j.

To derive the analytical approximation, we use the no-capital model of Alvarez and Lucas (2007). In this model, equilibrium imposes trade balance in every country. For country i,

$$L_i w_i = \sum_{k=1}^{n} L_k w_k D_{ki}, \tag{15}$$

where L_i denotes labor in efficiency units, w_i is the wage rate, and D_{ki} is the share of country k's spending on intermediate goods produced in country i. The left-hand side is country i's labor income, while the right-hand side is the total value of sales received by country i from all destination markets. Trade shares are given by

$$D_{ki} = (AB)^{-1/\theta} \left(\frac{w_i^{\beta} p_{mi}^{1-\beta} \tau_{ki}}{p_{mk}} \right)^{-1/\theta} \lambda_i,$$

where $\tau_{ki} \geq 1$ measures iceberg trade costs of shipping from i to k (i.e. τ_{ki} units must be shipped for one unit to arrive). The share of labor in the production of intermediates is β. Productivity draws are i.i.d. from an exponential distribution with λ_i the parameter that governs average productivity draws and $1/\theta$, common to all countries, the parameter that determines their dispersion. A and B are functions of parameters, and p_{mi} is the intermediate-goods price index that satisfies

$$p_{mi}^{-1/\theta} = (AB)^{-1/\theta} \sum_{k=1}^{n} \left(w_k^{\beta} p_{mk}^{1-\beta} \tau_{ik} \right)^{-1/\theta} \lambda_k. \tag{16}$$

Thus a country's technology parameter affects both its ability to sell abroad and the price indexes of countries that buy from it. Once wages and intermediate-goods prices are

determined, the final-good price is

$$p_i = \gamma^{-\gamma}(1-\gamma)^{-(1-\gamma)} w_i^{\gamma} p_{mi}^{1-\gamma}, \tag{17}$$

where γ is the share of labor in the production of the final good. Welfare per efficiency unit is therefore

$$c_i = \frac{w_i}{p_i}.$$

From (17), what matters for welfare is the movement in wages relative to the intermediate-goods price index.

A crisis in country j is modeled as a fall in its technology parameter,

$$\lambda_j \mapsto \lambda_j(1-\epsilon_j), \qquad \epsilon_j > 0.$$

A bailout prevents this fall in productivity. The first-order welfare gain from a bailout of country j for country i is therefore the avoided welfare loss,

$$u_{ij}^{+} \equiv -\frac{d}{d\epsilon_j}\left(\frac{w_i}{p_i}\right)\epsilon_j.$$

All variables and derivatives below are evaluated at the pre-crisis equilibrium, that is, at $\epsilon_j = 0$. The approximation keeps wages and intermediate-goods prices in all countries $k \notin \{i, j\}$ fixed, and solves for the induced first-order movements in (w_j, p_{mj}) and then in (w_i, p_{mi}).

For notational convenience, define

$$\psi_{ik} = \left(w_k^{\beta} p_{mk}^{1-\beta} \tau_{ik}\right)^{-1/\theta} \lambda_k,$$

with the term for source country j replaced by $\psi_{ij}(1-\epsilon_j)$ after the shock. At the initial equilibrium,

$$D_{ik} = \frac{\psi_{ik}}{\sum_{\ell=1}^{n} \psi_{i\ell}}. \tag{18}$$

Finally, define the local adjustment term, Δ_r

$$\Delta_r \equiv \beta(1+D_{rr}) + \theta\left[1-(1-\beta)D_{rr}\right], \qquad r \in \{i, j\}.$$

This term summarizes how wages and intermediate-goods prices jointly adjust in country r as a function of the domestic trade share D_{rr}. We focus on the empirically relevant case

$D_{rr} < 1$, so that the local linear systems below are non-singular.[27]

The effect of the shock on country j. We first compute the response of the shocked country. Let

$$x_j \equiv \frac{dw_j}{d\epsilon_j}, \qquad z_j \equiv \frac{w_j}{p_{mj}}\frac{dp_{mj}}{d\epsilon_j}.$$

The variable z_j is a wage-scaled change in the intermediate-goods price index. With this notation, the welfare response of country j depends on the difference $x_j - z_j$.

Differentiating the trade-balance condition (15) for country j gives

$$0 = L_j x_j - L_j x_j D_{jj} - \sum_{k=1}^{n} L_k w_k \left(\frac{\partial D_{kj}}{\partial w_j} x_j + \frac{\partial D_{kj}}{\partial p_{mj}} \frac{dp_{mj}}{d\epsilon_j} + \frac{\partial D_{kj}}{\partial \epsilon_j} \right). \tag{19}$$

The last term is the direct effect of the productivity shock on goods produced in country j, $\frac{\partial D_{kj}}{\partial \epsilon_j} = -D_{kj}$.[28] Using trade balance (15) in the initial equilibrium, $\sum_{k=1}^{n} \frac{L_k w_k}{L_j w_j} D_{kj} = 1$, together with

$$\begin{aligned}
\frac{\partial D_{kj}}{\partial w_j} &= -\frac{\beta}{\theta}\frac{D_{kj}}{w_j}, \\
\frac{\partial D_{kj}}{\partial p_{mj}} &= -\frac{1-\beta}{\theta}\frac{D_{kj}}{p_{mj}}, \qquad k \neq j, \\
\frac{\partial D_{jj}}{\partial p_{mj}} &= -\frac{1-\beta}{\theta}\frac{D_{jj}}{p_{mj}} + \frac{1}{\theta}\frac{D_{jj}}{p_{mj}} = \frac{\beta}{\theta}\frac{D_{jj}}{p_{mj}},
\end{aligned}$$

Equation (19) becomes, after multiplying by θ,

$$\left[\beta + \theta(1 - D_{jj})\right] x_j + (1 - \beta - D_{jj}) z_j = -\theta w_j. \tag{20}$$

The second equation to determine x_j and z_j comes from (16), the intermediate-goods price index in country j. Since the term associated with source country j is directly affected by the shock,

$$\frac{\partial \psi_{jj}}{\partial w_j} = -\frac{\beta}{\theta}\frac{\psi_{jj}}{w_j}, \qquad \frac{\partial \psi_{jj}}{\partial p_{mj}} = -\frac{1-\beta}{\theta}\frac{\psi_{jj}}{p_{mj}}, \qquad \frac{\partial \psi_{jj}}{\partial \epsilon_j} = -\psi_{jj}.$$

[27] The limiting case $D_{rr} = 1$ corresponds to a closed-economy normalization and would have to be treated separately.

[28] This is a partial derivative holding wages and price indexes fixed. The induced movement in the relevant price index is accounted for separately through the differentiated price-index equation. Economically, it captures the immediate fall in the competitiveness of country j's goods in every destination market.

Dividing the differentiated price-index equation by $p_{mj}^{-1/\theta}$ and using (18), we obtain, after multiplying by θw_j,

$$\beta D_{jj} x_j - [1 - (1-\beta) D_{jj}] \, z_j = -\theta w_j D_{jj}. \tag{21}$$

This equation captures the effect of the shock on the cost of the intermediate bundle in country j. Combining (20) and (21), we have

$$\begin{pmatrix} \beta + \theta(1 - D_{jj}) & 1 - \beta - D_{jj} \\ \beta D_{jj} & -[1 - (1-\beta) D_{jj}] \end{pmatrix} \begin{pmatrix} x_j \\ z_j \end{pmatrix} = \begin{pmatrix} -\theta w_j \\ -\theta w_j D_{jj} \end{pmatrix}.$$

The determinant of the coefficient matrix is

$$-(1 - D_{jj}) \{\beta(1 + D_{jj}) + \theta [1 - (1-\beta) D_{jj}]\} = -(1 - D_{jj}) \Delta_j.$$

Cramer's rule gives

$$x_j = -\frac{\theta w_j (1 + D_{jj})}{\Delta_j}, \tag{22}$$

$$z_j = \frac{\theta^2 w_j D_{jj}}{\Delta_j}. \tag{23}$$

Thus the shock lowers the wage in country j and raises its intermediate-goods price index. Both movements reduce real consumption in country j:

$$\frac{d}{d\epsilon_j} \left(\frac{w_j}{p_j} \right) = \frac{1 - \gamma}{p_j} (x_j - z_j) = -\frac{1 - \gamma}{p_j} \frac{\theta w_j \left[1 + (1 + \theta) D_{jj} \right]}{\Delta_j}.$$

Therefore, the first-order welfare gain to country j from avoiding its own productivity shock is

$$u_{jj}^{+} = \frac{1 - \gamma}{p_j} \frac{\theta w_j \left[1 + (1 + \theta) D_{jj} \right]}{\Delta_j} \epsilon_j. \tag{24}$$

The effect of the shock on a different country $i \neq j$. We now compute the effect on a trading partner i. The approximation allows the shock in j to affect country i through the induced changes in w_j and p_{mj}, together with the direct deterioration in the productivity of goods that i imports from j. Let

$$x_i \equiv \frac{dw_i}{d\epsilon_j}, \qquad z_i \equiv \frac{w_i}{p_{mi}} \frac{dp_{mi}}{d\epsilon_j}.$$

As before, the welfare effect for country i is determined by $x_i - z_i$.

Differentiating country i's trade-balance condition (15) and simplifying yields

$$\left[1 - D_{ii} + \frac{\beta}{\theta}\right] x_i + \left[\frac{1 - \beta - D_{ii}}{\theta}\right] z_i = \frac{L_j}{L_i} D_{ji} \left(x_j + \frac{1}{\theta} z_j\right).$$

The right-hand side is the bilateral export-demand channel. A change in country j's income and price index affects country i in proportion to D_{ji}, the share of country j's spending devoted to goods produced by country i. Multiplying by θ and using (22) and (23) (such that $\theta x_j + z_j = -\frac{\theta^2 w_j}{\Delta_j}$), gives

$$\left[\beta + \theta(1 - D_{ii})\right] x_i + (1 - \beta - D_{ii}) z_i = -\frac{L_j}{L_i} \frac{\theta^2 w_j D_{ji}}{\Delta_j}. \tag{25}$$

The second equation to solve for x_i and z_i comes from (16), the intermediate-goods price index in country i. The term associated with source country j changes directly with ϵ_j and indirectly through w_j and p_{mj}. Differentiating the price-index equation gives

$$\frac{\beta}{\theta} \frac{D_{ii}}{w_i} x_i + \left[\frac{1 - \beta}{\theta} \frac{D_{ii}}{w_i} - \frac{1}{\theta} \frac{1}{w_i}\right] z_i = -D_{ij} \left[1 + \frac{\beta}{\theta} \frac{1}{w_j} x_j + \frac{1 - \beta}{\theta} \frac{1}{w_j} z_j\right]. \tag{26}$$

The last three terms measure the import-cost channel. They are proportional to D_{ij}, the share of country i's intermediate spending devoted to goods from country j. Multiplying by θw_i and using (22) and (23) for x_j and z_j,[29]

$$\beta D_{ii} x_i - \left[1 - (1 - \beta) D_{ii}\right] z_i = -\frac{\theta^2 w_i D_{ij}}{\Delta_j}. \tag{27}$$

Equations (25) and (27) form the country-i linear system. To solve it, write

$$a_{ij} \equiv \frac{L_j}{L_i} \frac{\theta^2 w_j D_{ji}}{\Delta_j}, \qquad b_{ij} \equiv \frac{\theta^2 w_i D_{ij}}{\Delta_j}.$$

Then

$$\begin{pmatrix} \beta + \theta(1 - D_{ii}) & 1 - \beta - D_{ii} \\ \beta D_{ii} & -\left[1 - (1 - \beta) D_{ii}\right] \end{pmatrix} \begin{pmatrix} x_i \\ z_i \end{pmatrix} = \begin{pmatrix} -a_{ij} \\ -b_{ij} \end{pmatrix},$$

[29] This simplifies considerably the term in brackets on the RHS of (26)

$$\theta w_i + \beta \frac{w_i}{w_j} x_j + (1 - \beta) \frac{w_i}{w_j} z_j = \theta w_i \left[1 + \frac{-\beta(1 + D_{jj}) + (1 - \beta)\theta D_{jj}}{\Delta_j}\right] = \frac{\theta^2 w_i}{\Delta_j}.$$

where the final equality uses the definition of Δ_j.

with determinant $-(1-D_{ii})\Delta_i$. Cramer's rule gives

$$\begin{aligned} x_i &= \frac{b_{ij}(D_{ii}+\beta-1)-a_{ij}\left[1-(1-\beta)D_{ii}\right]}{(1-D_{ii})\Delta_i}, \\ z_i &= \frac{b_{ij}\left[\beta+\theta(1-D_{ii})\right]-a_{ij}\beta D_{ii}}{(1-D_{ii})\Delta_i}. \end{aligned}$$

Substituting back for a_{ij} and b_{ij},

$$\begin{aligned} \frac{dw_i}{d\epsilon_j} &= \frac{\theta^2}{(1-D_{ii})\Delta_i\Delta_j}\left[w_iD_{ij}(D_{ii}+\beta-1)-\frac{L_j}{L_i}w_jD_{ji}\left[1-(1-\beta)D_{ii}\right]\right], \\ \frac{w_i}{p_{mi}}\frac{dp_{mi}}{d\epsilon_j} &= \frac{\theta^2}{(1-D_{ii})\Delta_i\Delta_j}\left[w_iD_{ij}\left[\beta+\theta(1-D_{ii})\right]-\frac{L_j}{L_i}\beta D_{ii}w_jD_{ji}\right]. \end{aligned}$$

The separate wage and price-index responses depend on the domestic trade share D_{ii}. In the welfare effect, however, the terms involving $1-D_{ii}$ cancel:

$$x_i - z_i = -\frac{\theta^2}{\Delta_i\Delta_j}\left[\frac{L_j}{L_i}w_jD_{ji}+(1+\theta)w_iD_{ij}\right].$$

This expression is the key mechanism in the approximation. The first term is the export-demand channel: country i loses when country j's crisis reduces demand for goods produced in i. The second term is the import-cost channel: country i loses when goods produced in j become less productive and therefore more costly inside country i's intermediate bundle. Given that $\frac{d}{d\epsilon_j}\left(\frac{w_i}{p_i}\right)=\frac{1-\gamma}{p_i}(x_i-z_i)$, for $i\neq j$,

$$u_{ij}^{+}=\frac{1-\gamma}{L_ip_i}\frac{\theta^2\left[L_jw_jD_{ji}+(1+\theta)L_iw_iD_{ij}\right]}{\Delta_i\Delta_j}\epsilon_j, \qquad i\neq j.$$

The term $L_jw_jD_{ji}$ is the value of country i's sales to country j, while $L_iw_iD_{ij}$ is the value of country i's purchases from country j. In this approximation, both bilateral trade links raise the value to country i of preventing the shock in country j.

Multiplying by the population weight N_i, and combining the own-country (24) and cross-country cases, the approximate first-order bailout gains are

$$N_iu_{ij}^{+}=\begin{cases}\dfrac{N_j}{L_j}\dfrac{1-\gamma}{p_j}\dfrac{\theta L_jw_j\left[1+(1+\theta)D_{jj}\right]}{\Delta_j}\epsilon_j, & i=j, \\ \dfrac{N_i}{L_i}\dfrac{1-\gamma}{p_i}\dfrac{\theta^2\left[L_jw_jD_{ji}+(1+\theta)L_iw_iD_{ij}\right]}{\Delta_i\Delta_j}\epsilon_j, & i\neq j.\end{cases} \tag{28}$$

Equation (28) summarizes the main bilateral mechanisms that also discipline the full quantitative model. A country values a bailout of a trading partner because the bailout preserves demand for its exports and prevents an increase in the effective cost of imported intermediates. This helps explain why the full model delivers overwhelmingly positive values of u_{ij}^{+}: direct bilateral exposure to the crisis country creates positive welfare gains from avoiding the crisis.

We evaluate the approximation against the numerical benchmark generated by the full model. For the off-diagonal entries, $i \neq j$, which capture the cross-country spillovers that the approximation is designed to isolate, the approximation evaluated at $\epsilon_\lambda = 0.10$, where ϵ_λ denotes the proportional TFP shock used to approximate the 10 percent capital shock, has a correlation of 0.9986 with the full-model benchmark. Choosing the single value of ϵ_λ that minimizes off-diagonal squared error yields $\epsilon_\lambda = 0.0945$ and reduces off-diagonal RMSE by 32.1 percent relative to $\epsilon_\lambda = 0.10$. Because this common rescaling cancels when contribution shares are normalized, it affects bilateral spillover levels but not the contribution-share comparison below.

A more direct validation for governance asks whether the approximation reproduces the normalized contribution shares implied by the numerical model. For each regime at the baseline threshold, we hold that regime's full-model approval and participation sets fixed and replace only its numerical gross-gain matrix with the analytical counterpart. The correlation between analytical and numerical contribution shares is 0.99997 in the balanced regime and 0.99827 in the imbalanced regime, with mean absolute errors of 0.009 and 0.038 percentage points. For the change from balanced to imbalanced crisis risk, the correlation is 0.99906, the approximation predicts the direction of the change for all 127 countries, and the mean absolute error is 0.034 percentage points. The approximation therefore closely reproduces both the cross-country distribution of contribution shares and their reallocation as crisis risk becomes asymmetric.

At the numerical fragmented assignment, the same fixed-set comparison yields a correlation of 0.99850 and a mean absolute error of 0.036 percentage points. Analytical contribution shares aggregate to 42.06 percent for the US sphere and 57.94 percent for the China sphere, compared with 42.20 and 57.80 percent in the full numerical model. The quantitative results continue to be computed from the full numerical model.

A.2 Approximate assignment rule to geoeconomic spheres

Let US denote the United States and U the set of countries allied to the United States (but not the United States itself). Similarly CH denotes China and C its sphere of

influence.

To capture the additional trade costs from geoeconomic fragmentation, direct trade with the United States (China) is multiplied by τ_U (τ_C) for countries in the China (US) sphere. Trade with ordinary members of the opposing sphere is multiplied by $1+\zeta(\tau_U-1)$ or $1+\zeta(\tau_C-1)$, respectively. Under the symmetric calibration in appendix A.3, $\tau_U = \tau_C = \tau_H = \kappa_H$ and $1+\zeta(\tau_H-1) = \kappa_A$. The first-order rule below evaluates derivatives at the unfragmented baseline, $\tau_U = \tau_C = 1$. At an arbitrary τ_U, the direct derivative coefficients would be $1/\tau_U$ for the hegemon and $\zeta/[1+\zeta(\tau_U-1)]$ for ordinary sphere members. Define the local responses

$$x_i \equiv \left.\frac{dw_i}{d\tau_U}\right|_{\tau_U=1}, \qquad z_i \equiv \left.\frac{w_i}{p_{mi}}\frac{dp_{mi}}{d\tau_U}\right|_{\tau_U=1}.$$

With this notation, x_i is the wage response and z_i is the wage-scaled response of the intermediate-goods price index when country i joins the China sphere and faces the US-sphere wedges.

The direct partial effect of the wedge on trade shares is

$$\left.\frac{\partial D_{ki}}{\partial \tau_U}\right|_{\tau_U=1} = -\left(\frac{1}{\theta}D_{ki}(\zeta+(1-\zeta)\mathbb{1}_{k=US})\right)\mathbb{1}_{k\in\{US\}\cup U},$$

for exports of country i to destination k, and

$$\left.\frac{\partial D_{ik}}{\partial \tau_U}\right|_{\tau_U=1} = -\left(\frac{1}{\theta}D_{ik}(\zeta+(1-\zeta)\mathbb{1}_{k=US})\right)\mathbb{1}_{k\in\{US\}\cup U},$$

for imports of country i from source k.

We now differentiate (15) with respect to τ_U, keeping wages and intermediate-goods prices of countries $k \neq i$ fixed. Multiplying by θ and rearranging this derivative gives

$$L_i\left[\beta+\theta(1-D_{ii})\right]x_i + L_i\left(1-\beta-D_{ii}\right)z_i = -L_{US}w_{US}D_{USi} - \zeta\sum_{k\in U}L_k w_k D_{ki}. \tag{29}$$

The right-hand side is country i's export exposure to the US-centered bloc. Now we differentiate (16) with respect to τ_U, again keeping wages and intermediate-goods prices of countries $k \neq i$ fixed. This gives

$$\beta D_{ii}x_i - \left[1-(1-\beta)D_{ii}\right]z_i = -w_i\left(D_{iUS} + \zeta\sum_{k\in U}D_{ik}\right). \tag{30}$$

This is the second equation in a two-equation system for x_i and z_i. The right-hand side is country i's import exposure to the US-centered bloc.

To solve this system, we exploit its similarity to the productivity-shock system derived above. Define export exposure per unit of labor and import exposure as

$$s_i^X \equiv \frac{L_{US} w_{US} D_{USi} + \zeta \sum_{k \in U} L_k w_k D_{ki}}{L_i}, \qquad S_i^M \equiv w_i \left(D_{iUS} + \zeta \sum_{k \in U} D_{ik} \right).$$

Then (29) and (30) become

$$\begin{pmatrix} \beta + \theta(1 - D_{ii}) & 1 - \beta - D_{ii} \\ \beta D_{ii} & -\left[1 - (1-\beta) D_{ii}\right] \end{pmatrix} \begin{pmatrix} x_i \\ z_i \end{pmatrix} = \begin{pmatrix} -s_i^X \\ -S_i^M \end{pmatrix},$$

Using Cramer's rule, the solution is

$$\begin{aligned} x_i &= \frac{(\beta + D_{ii} - 1) S_i^M - \left[1 - (1-\beta) D_{ii}\right] s_i^X}{(1 - D_{ii}) \Delta_i}, \\ z_i &= \frac{\left[\theta(1 - D_{ii}) + \beta\right] S_i^M - \beta D_{ii} s_i^X}{(1 - D_{ii}) \Delta_i}. \end{aligned}$$

As with productivity shocks, the real-income effect of geoeconomic alignment is summarized by $x_i - z_i$:

$$x_i - z_i = -\frac{(1+\theta) w_i \left(D_{iUS} + \zeta \sum_{k \in U} D_{ik} \right) + \dfrac{1}{L_i} \left(L_{US} w_{US} D_{USi} + \zeta \sum_{k \in U} L_k w_k D_{ki} \right)}{\Delta_i}.$$

Thus, the welfare loss from a US-bloc trade-cost wedge is increasing in two sufficient statistics: country i's imports from the US-centered bloc and its exports to that bloc.

The same derivation applies to a China trade-cost wedge. Because the denominator is unchanged, comparing the two expressions yields the alignment rule in (7).

Table 2: Main calibration, $\delta = 0.45\%$, $\epsilon = 0.1$, majority $= 0.7$

Imbalanced					Balanced			
Country	Participant	Saved	w^*	Category	Country	Participant	Saved	w^*
CHN	Yes	Yes	0.1432	IG	CHN	Yes	Yes	0.2491
RUS	Yes	Yes	0.0871	HY	USA	Yes	Yes	0.0869
IDN	Yes	Yes	0.0779	HY	IND	Yes	Yes	0.0842
BRA	Yes	Yes	0.0725	HY	RUS	Yes	Yes	0.0389
USA	Yes	Yes	0.0493	IG	JPN	Yes	Yes	0.0389
IND	Yes	Yes	0.0485	IG	IDN	Yes	Yes	0.0350
NGA	Yes	Yes	0.0483	HY	DEU	Yes	Yes	0.0326
BGD	Yes	Yes	0.0288	HY	BRA	Yes	Yes	0.0322
VNM	Yes	Yes	0.0256	HY	NGA	Yes	Yes	0.0214
JPN	Yes	Yes	0.0220	IG	FRA	Yes	Yes	0.0212
TUR	Yes	Yes	0.0202	HY	MEX	Yes	Yes	0.0196
ARG	Yes	Yes	0.0197	HY	GBR	Yes	Yes	0.0186
DEU	Yes	Yes	0.0186	IG	ITA	Yes	Yes	0.0172
PAK	Yes	Yes	0.0165	HY	KOR	Yes	Yes	0.0164
UKR	Yes	Yes	0.0139	HY	BGD	Yes	Yes	0.0129
FRA	Yes	Yes	0.0120	IG	VNM	Yes	Yes	0.0121
MEX	Yes	Yes	0.0111	IG	CAN	Yes	Yes	0.0121
EGY	Yes	Yes	0.0109	HY	THA	Yes	Yes	0.0114
GBR	Yes	Yes	0.0106	IG	ESP	Yes	Yes	0.0111
ITA	Yes	Yes	0.0098	IG	SAU	Yes	Yes	0.0101
KOR	Yes	Yes	0.0093	IG	TUR	Yes	Yes	0.0091
IRQ	Yes	Yes	0.0088	HY	ARG	Yes	Yes	0.0087
ETH	Yes	Yes	0.0087	HY	PHL	Yes	Yes	0.0087
MMR	Yes	Yes	0.0085	HY	AUS	Yes	Yes	0.0086
AGO	Yes	Yes	0.0076	HY	PAK	Yes	Yes	0.0075
CAN	Yes	Yes	0.0068	IG	POL	Yes	Yes	0.0072
THA	Yes	Yes	0.0066	IG	NLD	Yes	Yes	0.0067
ESP	Yes	Yes	0.0065	IG	UKR	Yes	Yes	0.0062
UZB	Yes	Yes	0.0063	HY	MYS	Yes	Yes	0.0058
SAU	Yes	Yes	0.0057	IG	COL	Yes	Yes	0.0057
ECU	Yes	Yes	0.0057	HY	PER	Yes	Yes	0.0053
PRT	Yes	Yes	0.0052	HY	CHE	Yes	Yes	0.0049

Table 3: Main calibration, $\delta = 0.45\%$, $\epsilon = 0.1$, majority $= 0.7$ (continued)

Imbalanced					Balanced			
Country	Participant	Saved	w^*	Category	Country	Participant	Saved	w^*
TZA	Yes	Yes	0.0051	HY	EGY	Yes	Yes	0.0049
HUN	Yes	Yes	0.0050	HY	ZAF	Yes	Yes	0.0044
PHL	Yes	Yes	0.0049	IG	KAZ	Yes	Yes	0.0040
AUS	Yes	Yes	0.0048	IG	IRQ	Yes	Yes	0.0039
MAR	Yes	Yes	0.0046	HY	BEL	Yes	Yes	0.0039
GRC	Yes	Yes	0.0045	HY	MMR	Yes	Yes	0.0039
GHA	Yes	Yes	0.0044	HY	ETH	Yes	Yes	0.0038
POL	Yes	Yes	0.0042	IG	AUT	Yes	Yes	0.0038
KEN	Yes	Yes	0.0041	HY	SWE	Yes	Yes	0.0037
LKA	Yes	Yes	0.0040	HY	ARE	Yes	Yes	0.0034
NLD	Yes	Yes	0.0038	IG	AGO	Yes	Yes	0.0033
MYS	Yes	Yes	0.0036	IG	CZE	Yes	Yes	0.0033
BLR	Yes	Yes	0.0034	HY	ROU	Yes	Yes	0.0033
CIV	Yes	Yes	0.0033	HY	CHL	Yes	Yes	0.0031
COL	Yes	Yes	0.0033	IG	UZB	Yes	Yes	0.0028
GTM	Yes	Yes	0.0031	HY	IRL	Yes	Yes	0.0028
KHM	Yes	Yes	0.0031	HY	DNK	Yes	Yes	0.0025
PER	Yes	Yes	0.0030	IG	ECU	Yes	Yes	0.0025
DOM	Yes	Yes	0.0029	HY	NOR	Yes	Yes	0.0025
UGA	Yes	Yes	0.0029	HY	PRT	Yes	Yes	0.0024
BOL	Yes	Yes	0.0028	HY	HUN	Yes	Yes	0.0023
CHE	Yes	Yes	0.0028	IG	TZA	Yes	Yes	0.0022
NPL	Yes	Yes	0.0026	HY	QAT	Yes	Yes	0.0022
CMR	Yes	Yes	0.0025	HY	MAR	Yes	Yes	0.0021
ZAF	Yes	Yes	0.0025	IG	SGP	Yes	Yes	0.0021
SDN	Yes	Yes	0.0023	HY	FIN	Yes	Yes	0.0020
BGR	Yes	Yes	0.0023	HY	ISR	Yes	Yes	0.0020
KAZ	Yes	Yes	0.0023	IG	GRC	Yes	Yes	0.0020
BEL	Yes	Yes	0.0022	IG	GHA	Yes	Yes	0.0019
AUT	Yes	Yes	0.0021	IG	OMN	Yes	Yes	0.0019
SWE	Yes	Yes	0.0021	IG	KEN	Yes	Yes	0.0018
LAO	Yes	Yes	0.0021	HY	LKA	Yes	Yes	0.0018

Table 4: Main calibration, $\delta = 0.45\%$, $\epsilon = 0.1$, majority = 0.7 (continued)

Imbalanced					Balanced			
Country	Participant	Saved	w^*	Category	Country	Participant	Saved	w^*
MDG	Yes	Yes	0.0021	HY	BLR	Yes	Yes	0.0015
ARE	Yes	Yes	0.0019	IG	KWT	Yes	Yes	0.0015
ROU	Yes	Yes	0.0019	IG	CIV	Yes	Yes	0.0015
CZE	Yes	Yes	0.0019	IG	GTM	Yes	Yes	0.0014
SRB	Yes	Yes	0.0019	HY	KHM	Yes	Yes	0.0014
NER	Yes	Yes	0.0018	HY	DOM	Yes	Yes	0.0013
CHL	Yes	Yes	0.0018	IG	UGA	Yes	Yes	0.0013
ZWE	Yes	Yes	0.0017	HY	BOL	Yes	Yes	0.0013
TUN	Yes	Yes	0.0017	HY	AZE	Yes	Yes	0.0012
ZMB	Yes	Yes	0.0016	HY	NZL	Yes	Yes	0.0012
CRI	Yes	Yes	0.0016	HY	NPL	Yes	Yes	0.0012
HRV	Yes	Yes	0.0016	HY	SVK	Yes	Yes	0.0011
IRL	Yes	Yes	0.0016	IG	CMR	Yes	Yes	0.0011
PRY	Yes	Yes	0.0015	HY	BGR	Yes	Yes	0.0010
BFA	Yes	Yes	0.0015	HY	SDN	Yes	Yes	0.0010
DNK	Yes	Yes	0.0014	IG	LAO	Yes	Yes	9.41E-04
NOR	Yes	Yes	0.0014	IG	MDG	Yes	Yes	9.20E-04
MOZ	Yes	Yes	0.0014	HY	SRB	Yes	Yes	8.51E-04
MLI	Yes	Yes	0.0013	HY	NER	Yes	Yes	7.80E-04
SGP	Yes	Yes	0.0013	IG	TUN	Yes	Yes	7.76E-04
HTI	Yes	Yes	0.0013	HY	ZWE	Yes	Yes	7.66E-04
QAT	Yes	Yes	0.0013	IG	ZMB	Yes	Yes	7.23E-04
COG	Yes	Yes	0.0013	HY	HRV	Yes	Yes	7.18E-04
FIN	Yes	Yes	0.0012	IG	CRI	Yes	Yes	7.15E-04
ISR	Yes	Yes	0.0012	IG	LTU	Yes	Yes	6.80E-04
HND	Yes	Yes	0.0012	HY	PRY	Yes	Yes	6.76E-04
RWA	Yes	Yes	0.0011	HY	BFA	Yes	Yes	6.70E-04
SEN	Yes	Yes	0.0011	HY	URY	Yes	Yes	6.55E-04
JOR	Yes	Yes	0.0011	HY	MOZ	Yes	Yes	6.21E-04
OMN	Yes	Yes	0.0011	IG	SVN	Yes	Yes	5.85E-04
SLV	Yes	Yes	0.0010	HY	MLI	Yes	Yes	5.80E-04
MNG	Yes	Yes	9.41E-04	HY	HTI	Yes	Yes	5.67E-04

Table 5: Main calibration, $\delta = 0.45\%$, $\epsilon = 0.1$, majority $= 0.7$ (continued)

Imbalanced				Balanced			
Country	Participant	Saved	w^*	Category	Country	Participant	Saved	w^*
LBN	Yes	Yes	9.12E-04	HY	COG	Yes	Yes	5.55E-04
TJK	Yes	Yes	8.63E-04	HY	HND	Yes	Yes	5.17E-04
KWT	Yes	Yes	8.56E-04	IG	SEN	Yes	Yes	4.99E-04
NIC	Yes	Yes	7.80E-04	HY	JOR	Yes	Yes	4.98E-04
BEN	Yes	Yes	7.53E-04	HY	RWA	Yes	Yes	4.92E-04
ALB	Yes	Yes	7.53E-04	HY	SLV	Yes	Yes	4.64E-04
GNQ	Yes	Yes	7.41E-04	HY	PAN	Yes	Yes	4.58E-04
AZE	Yes	Yes	6.81E-04	IG	LVA	Yes	Yes	4.41E-04
NZL	Yes	Yes	6.68E-04	IG	MNG	Yes	Yes	4.19E-04
SVK	Yes	Yes	6.53E-04	IG	LBN	Yes	Yes	4.08E-04
MWI	Yes	Yes	6.53E-04	HY	TJK	Yes	Yes	3.83E-04
ARM	Yes	Yes	6.32E-04	HY	BHR	Yes	Yes	3.59E-04
GAB	Yes	Yes	6.05E-04	HY	BWA	Yes	Yes	3.55E-04
MDA	Yes	Yes	5.94E-04	HY	NIC	Yes	Yes	3.48E-04
GEO	Yes	Yes	5.48E-04	HY	ALB	Yes	Yes	3.34E-04
MKD	Yes	Yes	4.48E-04	HY	BEN	Yes	Yes	3.33E-04
JAM	Yes	Yes	4.48E-04	HY	GNQ	Yes	Yes	3.28E-04
BIH	Yes	Yes	4.24E-04	HY	EST	Yes	Yes	2.96E-04
LTU	Yes	Yes	4.05E-04	IG	MWI	Yes	Yes	2.89E-04
URY	Yes	Yes	3.86E-04	IG	ARM	Yes	Yes	2.80E-04
SVN	Yes	Yes	3.38E-04	IG	TTO	Yes	Yes	2.71E-04
LVA	Yes	Yes	2.65E-04	IG	GAB	Yes	Yes	2.68E-04
PAN	Yes	Yes	2.62E-04	IG	MDA	Yes	Yes	2.64E-04
BHR	Yes	Yes	2.04E-04	IG	GEO	Yes	Yes	2.44E-04
BWA	Yes	Yes	2.01E-04	IG	MKD	Yes	Yes	2.02E-04
EST	Yes	Yes	1.82E-04	IG	JAM	Yes	Yes	1.99E-04
TTO	Yes	Yes	1.54E-04	IG	BIH	Yes	Yes	1.91E-04
GNB	Yes	Yes	1.14E-04	HY	NAM	Yes	Yes	1.71E-04
NAM	Yes	Yes	9.67E-05	IG	MUS	Yes	Yes	1.24E-04
MUS	Yes	Yes	7.07E-05	IG	GMB	Yes	No	7.12E-07
GMB	Yes	No	6.04E-07	HY	GNB	Yes	No	4.47E-08

Table 6: Country-Level Spheres, Actual IMF Quotas, and Fragmented Contributions

ISO	Equilibrium sphere	Historical benchmark	Actual quota share	Model ω^*
AGO	China	US	0.1620	0.7593
ALB	US	China	0.0305	0.0754
ARE	China	China	0.5059	0.1905
ARG	China	China	0.6976	1.9755
ARM	US	China	0.0282	0.0633
AUS	China	US	1.4385	0.4850
AUT	US	US	0.8606	0.2148
AZE	US	China	0.0857	0.0682
BEL	US	US	1.4031	0.2187
BEN	US	US	0.0271	0.0754
BFA	US	China	0.0264	0.1518
BGD	China	China	0.2335	2.8912
BGR	US	US	0.1962	0.2294
BHR	China	US	0.0865	0.0203
BIH	US	China	0.0580	0.0425
BLR	US	China	0.1492	0.3399
BOL	China	US	0.0526	0.2854
BRA	China	China	2.4168	7.2626
BWA	China	US	0.0432	0.0201
CAN	US	US	2.4129	0.6838
CHE	US	US	1.2632	0.2770
CHL	China	US	0.3818	0.1754
CHN	China	China	6.6720	14.2626
CIV	US	US	0.1424	0.3340
CMR	US	China	0.0604	0.2535
COG	China	China	0.0355	0.1256
COL	US	US	0.4475	0.3258
CRI	US	US	0.0809	0.1614
CZE	US	US	0.4772	0.1900
DEU	US	US	5.8296	1.8574
DNK	US	US	0.7528	0.1441
DOM	US	US	0.1045	0.2894

A.3 Calibration of the fragmentation exercise

The fragmentation exercise contrasts the unbalanced economy without additional trade barriers with a severely divided world.

We choose the rival-hegemon iceberg multiplier so that its direct gravity effect halves trade. Since the trade-cost elasticity is $-1/\theta$, the two multipliers satisfy

$$\kappa_H^{-1/\theta} = 0.5, \qquad \kappa_A^{-1/\theta} = 0.5^{\rho},$$

giving, at $\theta = 0.1818$,

$$\kappa_H = 0.5^{-\theta} = 1.13, \qquad \kappa_A = 0.5^{-\theta\rho} = 1.10, \qquad \zeta = \frac{\kappa_A - 1}{\kappa_H - 1} = 0.73.$$

These are increases of 13.43 and 9.78 percent relative to unfragmented trade costs, corresponding to conditional trade declines of 50 and 40.14 percent. The 50 percent decline

is our chosen severe scenario, not an estimate from Broner et al. (2026). Costs are fixed before solving the model; realized trade also responds to equilibrium prices, incomes, and sphere choices.

Each directed bilateral pair receives exactly one fragmentation multiplier. Within-sphere and domestic costs are unchanged; trade with the rival hegemon receives κ_H; trade between ordinary countries in opposing spheres receives κ_A; and US–China trade receives κ_H once. The two wedges are not stacked. Applying the US-based alignment estimates symmetrically to both spheres is a modeling assumption. The same new multiplier applies in both directions, while the estimated pre-fragmentation cost matrix remains asymmetric.

A.4 Construction of the fragmentation counterfactual

The numerical exercise is constructed independently of the analytical approximation in (7). We begin with Panel B of the 2010–2019 treaty-alignment map in the 2025 version of Broner et al. (2025). The United States and China are fixed in their respective spheres. For the remaining countries, the map supplies an initial political classification. To capture the persistence of historically stronger US relationships, we apply an asymmetric trade override: a country classified in the China sphere is moved to the US starting sphere if its 2019 bilateral goods trade—imports plus exports—is larger with the United States than with China; a country classified in the US sphere remains there. On a trade tie, the map classification is retained. This procedure yields a predetermined historical benchmark of 77 countries in the US sphere and 50 in the China sphere.[30]

For assignment a, country i evaluates the model's full expected aggregate output under the additive arrival-intensity closure described above:

$$V_i(a) = L_i Y_i^0(a) - \sum_j \mu_j L_i \left[Y_i^0(a) - Y_i^j(a)\right] + G_i^{IMF}(a),$$

where $Y_i^j(a)$ is country i's real factor income per worker when country j experiences the capital shock and receives no intervention, and $G_i^{IMF}(a)$ is the incremental expected net gain from IMF intervention under the imbalanced crisis-intensity vector. The first two terms give expected aggregate output absent intervention; $G_i^{IMF}(a)$ adds the signed

[30]The map was transcribed independently three times. Ninety-one classifications were unanimous, 22 were resolved by a blinded majority recheck, and four close cases were adjudicated against labeled source-map locations. For ten observations marked NA or illegible, we apply the prespecified US-sphere default: entry into the China benchmark requires an affirmative China classification. All 125 non-hegemons are matched to BACI trade data, and no trade tie occurs. Appendix table 6 reports the historical and equilibrium classifications for every country.

welfare effect of preventing shocks in approved borrower states and subtracts country i's expected financing costs. For each candidate switch, we re-solve the no-crisis equilibrium, every borrower-specific capital-shock equilibrium, and the governance fixed point. Wages, prices, returns, trade flows, contribution shares, vote-specific weights, bailout decisions, and Fund participation therefore all adjust to the candidate assignment.

The assignment criterion therefore includes both channels through which fragmentation affects expected welfare: it changes output and the propagation of losses in every borrower state, and it changes the endogenous value and cost of IMF intervention. Crisis losses are neither omitted nor counted twice.

For each assignment, the governance calculation jointly determines the participant set P and the set S of approved borrower states. Contribution shares and vote-specific weights are recomputed as these sets adjust, and previously rejected borrower states can be reconsidered. Participation is checked against every country's unilateral entry or exit, recomputing governance for the alternative membership while holding that assignment's trade equilibrium fixed. Nonparticipants retain spillovers from other countries' rescues but neither finance, vote on, nor receive assistance.

The search starts from the historical assignment and considers countries in fixed descending order of 2019 nominal GDP. Following an accepted switch, it restarts at the top of that order. We then verify the reported 75–52 assignment independently by evaluating every non-hegemon's unilateral sphere deviation, allowing the trade equilibrium and governance outcomes to adjust. The resulting assignment is locally stable; this procedure does not establish uniqueness. The baseline uses the 70 percent voting threshold. For the 50 percent robustness exercise, we recompute governance at the same assignment and repeat the complete unilateral-deviation check.

The fragmented result in table 12 is conditional on the calibrated trade wedges, the level and distribution of crisis-arrival intensities, the governance closure, and the full expected-output assignment objective. The complete deviation scan establishes local stability of the reported assignment, not uniqueness; other locally stable assignments may exist.

A.5 Additional results for the fragmentation exercise

The fragmentation exercise combines the imbalanced crisis intensities with the trade-cost multipliers $\kappa_H = 1.13$ and $\kappa_A = 1.10$. This appendix reports the assignment, trade effects, contribution shares, and sensitivity to the voting threshold.

Assignment and trade. The reported locally stable assignment places 75 countries in the US sphere and 52 in the China sphere. Its classification differs from the 77–50 historical starting assignment for 38 countries: 20 join the China sphere, including Australia, India, Saudi Arabia, and Singapore, while 18 join the US sphere, including Nigeria, Russia, Spain, and Turkey. The final grouping reflects countries' economic incentives under severe fragmentation rather than an imposed treaty classification. No unilateral switch by any of the 125 non-hegemon countries raises full expected output at the reported assignment. At the baseline 70 percent threshold, 126 countries participate and receive assistance in their respective crisis states; The Gambia remains outside the Fund and receives no bailout.

The additional barriers substantially redirect trade. Relative to the unfragmented economy, nominal intermediate trade with the opposing hegemon falls by 48.47 percent, and trade between ordinary members of opposing spheres falls by 38.54 percent. Cross-sphere trade falls by 45.39 percent overall, while international trade within spheres increases by 2.41 percent. Aggregate international trade declines by 2.52 percent. Appendix table 13 reports these changes using a common wage numeraire and the same country-pair classifications in both economies.

Contribution shares and existing quota gaps. Despite this substantial trade reallocation, contribution shares remain close to the unbalanced benchmark. The high-yield aggregate rises from 25.74 percent in the balanced regime to 57.52 percent in the unbalanced regime and 57.59 percent under fragmentation. Across all 127 countries, the largest change from the unbalanced to the fragmented regime is China's decline of 0.060 percentage points, from 14.32 to 14.26 percent; the mean absolute change is 0.0016 points. The main redistribution therefore comes from heterogeneous crisis risk, not from the additional trade barriers.

Expected benefits from a country's own rescue help explain this result. In the fragmented equilibrium, they account for 95.53 percent of China's gross expected benefit and 97.65 percent of the US gross expected benefit. For these two countries, changes in foreign spillovers affect a relatively small component of the total. Large changes in bilateral trade can therefore coexist with small changes in contribution shares.

For completeness, table 12 aggregates actual quotas and contribution shares over the countries in the reported fragmented assignment. All columns use these same country groups. This comparison distinguishes a pre-existing quota gap from any additional effect of fragmentation on their combined contribution shares.

The US-sphere countries hold 65.3 percent of sample quota shares and account for 42.2 percent of fragmented ω^*; the China-sphere countries hold 34.7 percent of quotas

and account for 57.8 percent of ω^*. The latter group's quota-to-contribution gap is 23.1 percentage points. Its combined contribution benchmark is already 57.8070 percent in the unfragmented imbalanced economy and becomes 57.8015 percent under fragmentation, a change of only 0.0055 percentage points. The large gap between actual quotas and the benchmark therefore predates the additional trade barriers.

These aggregates reinforce the country-level result: severe trade segmentation leaves the post-Bretton-Woods quota mismatch largely unchanged. They do not establish that economic spheres would act as political coalitions within the Fund or choose to form alternative lenders. Table 6 reports actual quotas, sphere classifications, and fragmented-model contribution shares for all 127 countries.

Trade reallocation. Table 13 reports changes in the no-crisis nominal intermediate-trade index. We classify country pairs using the 75–52 equilibrium assignment in both economies, so the comparison follows the same trading relationships before and after imposing fragmentation costs. The index is constructed as $F_{ij} = m_{ij}L_j w_j$, where m_{ij} is the share of importer j's intermediate expenditure allocated to exporter i, using a common wage numeraire. These are nominal trade changes, not changes in physical trade volumes.

Voting threshold. The baseline threshold is 70 percent. We also solve governance under the 50 percent rule in (2), holding the 75–52 sphere assignment fixed, and then evaluate every non-hegemon's unilateral sphere deviation under that rule. The assignment remains locally stable. This exercise changes the voting requirement, not the fragmentation costs.

At the 70 percent threshold, 126 countries participate and their borrower states are approved; The Gambia remains outside the Fund. At 50 percent, all 127 countries participate and all borrower states are approved. The China sphere's aggregate contribution share is 57.8015 percent at 70 percent and 57.7978 percent at 50 percent, against an actual quota share of 34.7194 percent. Table 14 reports the country-level comparison for the G20 sovereign members and Spain.

Table 7: Country-Level Spheres, Actual IMF Quotas, and Fragmented Contributions (continued)

ISO	Equilibrium sphere	Historical benchmark	Actual quota share	Model ω^*
ECU	US	US	0.1527	0.5714
EGY	US	China	0.4459	1.0919
ESP	US	China	2.0871	0.6453
EST	US	US	0.0533	0.0182
ETH	China	China	0.0658	0.8688
FIN	US	US	0.5276	0.1188
FRA	US	US	4.4115	1.2027
GAB	US	China	0.0473	0.0607
GBR	US	US	4.4115	1.0577
GEO	US	US	0.0461	0.0548
GHA	US	US	0.1615	0.4387
GMB	US	US	0.0136	0.0000
GNB	US	China	0.0062	0.0115
GNQ	US	US	0.0345	0.0743
GRC	US	US	0.5316	0.4479
GTM	US	US	0.0938	0.3154
HND	US	US	0.0547	0.1162
HRV	US	US	0.1570	0.1601
HTI	US	US	0.0359	0.1283
HUN	US	China	0.4246	0.5006
IDN	China	China	1.0174	7.8001
IND	China	US	2.8704	4.8461
IRL	US	US	0.7551	0.1575
IRQ	US	US	0.3642	0.8802
ISR	US	US	0.4204	0.1161
ITA	US	US	3.2985	0.9768
JAM	US	US	0.0838	0.0449
JOR	US	US	0.0751	0.1096
JPN	China	China	6.7459	2.2016
KAZ	China	China	0.2535	0.2267
KEN	China	China	0.1188	0.4073
KHM	China	China	0.0383	0.3073

Table 8: Country-Level Spheres, Actual IMF Quotas, and Fragmented Contributions (continued)

ISO	Equilibrium sphere	Historical benchmark	Actual quota share	Model ω^*
KOR	China	China	1.8785	0.9269
KWT	China	China	0.4232	0.0857
LAO	China	China	0.0232	0.2125
LBN	US	China	0.1387	0.0910
LKA	China	China	0.1267	0.4049
LTU	US	US	0.0967	0.0405
LVA	US	US	0.0727	0.0265
MAR	US	US	0.1958	0.4645
MDA	US	US	0.0378	0.0595
MDG	China	US	0.0535	0.2073
MEX	US	US	1.9508	1.1098
MKD	US	US	0.0307	0.0448
MLI	US	US	0.0408	0.1314
MMR	China	China	0.1131	0.8536
MNG	China	China	0.0158	0.0943
MOZ	China	US	0.0497	0.1347
MUS	China	US	0.0311	0.0070
MWI	China	US	0.0304	0.0654
MYS	China	China	0.7954	0.3570
NAM	China	US	0.0418	0.0097
NER	US	US	0.0288	0.1767
NGA	US	China	0.5372	4.8463
NIC	US	US	0.0569	0.0780
NLD	US	US	1.9122	0.3820
NOR	US	China	0.8218	0.1405
NPL	China	China	0.0343	0.2592
NZL	China	China	0.2741	0.0667
OMN	China	China	0.1192	0.1059
PAK	China	China	0.4445	1.6503
PAN	US	US	0.0825	0.0261
PER	China	China	0.2921	0.2999
PHL	China	US	0.4471	0.4927

Table 9: Country-Level Spheres, Actual IMF Quotas, and Fragmented Contributions (continued)

ISO	Equilibrium sphere	Historical benchmark	Actual quota share	Model ω^*
POL	US	US	0.8964	0.4168
PRT	US	US	0.4509	0.5211
PRY	China	China	0.0441	0.1526
QAT	China	US	0.1609	0.1264
ROU	US	US	0.3965	0.1916
RUS	US	China	2.8243	8.6970
RWA	China	US	0.0351	0.1115
SAU	China	US	2.1871	0.5748
SDN	US	China	0.1379	0.2318
SEN	US	US	0.0708	0.1106
SGP	China	US	0.8518	0.1287
SLV	US	US	0.0629	0.1045
SRB	US	US	0.1433	0.1891
SVK	US	US	0.2191	0.0652
SVN	US	US	0.1284	0.0338
SWE	US	US	0.9696	0.2124
THA	China	China	0.7030	0.6564
TJK	China	China	0.0381	0.0865
TTO	US	US	0.1028	0.0154
TUN	US	US	0.1193	0.1711
TUR	US	China	1.0197	2.0093
TZA	China	China	0.0871	0.5083
UGA	China	US	0.0790	0.2884
UKR	US	US	0.4403	1.3922
URY	China	China	0.0939	0.0387
USA	US	US	18.1654	4.9236
UZB	China	China	0.1206	0.6342
VNM	China	China	0.2524	2.5618
ZAF	China	China	0.6678	0.2505
ZMB	China	US	0.2141	0.1637
ZWE	China	US	0.1547	0.1734

Notes: All numerical entries are percentages. Actual quota shares are renormalized over the 127-country model sample. Model ω^* is the normalized contribution share in the reported locally stable fragmented equilibrium. The historical benchmark combines the treaty-map classification with the asymmetric 2019 trade override described in the text. The Gambia is outside the Fund and has a zero contribution share. Source: authors' calculations and International Monetary Fund (2026b).

Table 10: Robustness: $\delta = 0.58\%$, $\epsilon = 0.1$, majority $= 0.7$

Country	ISO	Cat.	Imbalanced	Balanced	IMF quota
Argentina	ARG	HY	1.97	0.87	0.70
Australia	AUS	IG	0.48	0.86	1.44
Brazil	BRA	HY	7.25	3.22	2.42
Canada	CAN	IG	0.68	1.21	2.41
China	CHN	IG	14.32	24.91	6.67
France	FRA	IG	1.20	2.12	4.41
Germany	DEU	IG	1.86	3.26	5.83
India	IND	IG	4.85	8.42	2.87
Indonesia	IDN	HY	7.79	3.50	1.02
Italy	ITA	IG	0.98	1.72	3.30
Japan	JPN	IG	2.20	3.89	6.75
Korea	KOR	IG	0.93	1.64	1.88
Mexico	MEX	IG	1.11	1.96	1.95
Russian Federation	RUS	HY	8.71	3.89	2.82
Saudi Arabia	SAU	IG	0.57	1.01	2.19
South Africa	ZAF	IG	0.25	0.44	0.67
Spain	ESP	IG	0.65	1.11	2.09
Turkey	TUR	HY	2.02	0.91	1.02
United Kingdom	GBR	IG	1.06	1.86	4.41
United States	USA	IG	4.93	8.69	18.17

Notes: Model contribution weights are rescaled to percentages. Actual quota shares are renormalized over the 127-country model sample. Source: authors' calculations and International Monetary Fund (2026b).

Table 11: Robustness: General voting rule ($\gamma = 1/2$), $\delta = 0.45\%$, $\epsilon = 0.10$

Country	ISO	Cat.	Imbalanced	Balanced	IMF quota
Argentina	ARG	HY	1.97	0.87	0.70
Australia	AUS	IG	0.48	0.86	1.44
Brazil	BRA	HY	7.25	3.22	2.42
Canada	CAN	IG	0.68	1.21	2.41
China	CHN	IG	14.32	24.91	6.67
France	FRA	IG	1.20	2.12	4.41
Germany	DEU	IG	1.86	3.26	5.83
India	IND	IG	4.85	8.42	2.87
Indonesia	IDN	HY	7.79	3.50	1.02
Italy	ITA	IG	0.98	1.72	3.30
Japan	JPN	IG	2.20	3.89	6.75
Korea	KOR	IG	0.93	1.64	1.88
Mexico	MEX	IG	1.11	1.96	1.95
Russian Federation	RUS	HY	8.71	3.89	2.82
Saudi Arabia	SAU	IG	0.57	1.01	2.19
South Africa	ZAF	IG	0.25	0.44	0.67
Spain	ESP	IG	0.65	1.11	2.09
Turkey	TUR	HY	2.02	0.91	1.02
United Kingdom	GBR	IG	1.06	1.86	4.41
United States	USA	IG	4.93	8.69	18.17

Notes: Model contribution weights are normalized contribution shares, in percent. At a 50 percent threshold, the voting rule coincides with (2) up to ties; no exact ties occur. Actual quota shares are renormalized over the 127-country model sample. Source: authors' calculations and International Monetary Fund (2026b).

Table 12: Current IMF Quotas and Contribution Benchmarks by Economic Sphere

Sphere	Countries	Sample-normalized quota share	Balanced ω^*	Imbalanced ω^*	Fragmented ω^*
US	75	65.2806	39.6329	42.1930	42.1985
China	52	34.7194	60.3671	57.8070	57.8015

Notes: Entries are percentages. Actual quota shares are from the IMF member table updated on July 28, 2026, and renormalized over the 127-country model sample, which accounts for 95.91 percent of total IMF quotas. The model is calibrated to 2019; actual quotas provide the current policy benchmark. The 16th General Review approved an equiproportional quota increase that had not yet become effective as of July 2026 and will not by itself reallocate quota shares (International Monetary Fund, 2023, 2026a). All columns use the same 75–52 sphere classification. Balanced and imbalanced contributions are computed without the additional fragmentation costs; the fragmented column uses the sphere-dependent costs described in the text. The voting threshold is 70 percent. Model contribution shares sum to 100 percent, with zero shares for nonparticipants. Source: authors' calculations and International Monetary Fund (2026b).

Table 13: Trade under Fragmentation Relative to the Unfragmented Economy

Trade links	Percentage change
Opposing-hegemon links	-48.47
Ordinary cross-sphere links	-38.54
All cross-sphere links	-45.39
Within-sphere international links	2.41
All international links	-2.52

Notes: Entries are percentage changes in the no-crisis nominal intermediate-trade index, expressed in a common wage numeraire. Pair classifications are held fixed at the fragmented equilibrium assignment. Opposing-hegemon links include US–China trade; ordinary cross-sphere links exclude both hegemons. Domestic purchases are excluded. Source: authors' calculations.

Table 14: Fragmented Contribution Shares under Alternative Voting Thresholds

Country	ISO	70 percent threshold	50 percent threshold	Change	Sphere
Argentina	ARG	1.9755	1.9754	-0.0001	China
Australia	AUS	0.4850	0.4850	0.0000	China
Brazil	BRA	7.2626	7.2621	-0.0005	China
Canada	CAN	0.6838	0.6838	0.0000	US
China	CHN	14.2626	14.2617	-0.0009	China
France	FRA	1.2027	1.2026	-0.0001	US
Germany	DEU	1.8574	1.8573	-0.0001	US
India	IND	4.8461	4.8458	-0.0003	China
Indonesia	IDN	7.8001	7.7996	-0.0005	China
Italy	ITA	0.9768	0.9767	-0.0001	US
Japan	JPN	2.2016	2.2014	-0.0001	China
Korea	KOR	0.9269	0.9268	-0.0001	China
Mexico	MEX	1.1098	1.1097	-0.0001	US
Russian Federation	RUS	8.6970	8.6964	-0.0006	US
Saudi Arabia	SAU	0.5748	0.5748	0.0000	China
South Africa	ZAF	0.2505	0.2505	0.0000	China
Spain	ESP	0.6453	0.6452	0.0000	US
Turkey	TUR	2.0093	2.0092	-0.0001	US
United Kingdom	GBR	1.0577	1.0577	-0.0001	US
United States	USA	4.9236	4.9233	-0.0003	US

Notes: Contribution shares are percentages, normalized across the full model sample rather than the countries displayed. Changes are in percentage points and are computed before rounding. Both columns use the same fragmentation costs and the same 75–52 assignment, with governance recomputed at each voting threshold. *Sphere* denotes the fragmented-equilibrium assignment. Source: authors' calculations.